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(54) **METHOD AND APPARATUS RELATED TO DATA GENERATION FRAMEWORK**

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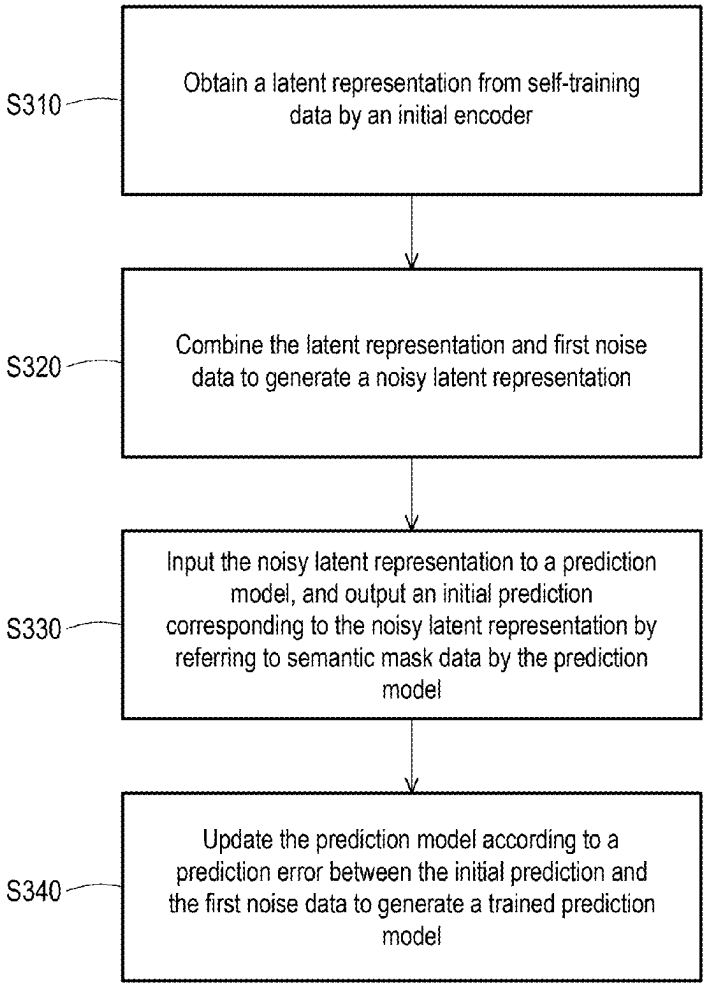
(57) **ABSTRACT**

A method and an apparatus related to a data generation framework are provided. A first code is obtained from reference data by a style encoder. A second code is obtained from latent encoding by a mapping network. The first code and the second code are both a style code, and the style code corresponds to one or more style options. First source data is input to a generator, and a first output corresponding to the first source data is output by referring to the style code by the generator. The first output is input to a discriminator, and a second output corresponding to the first output is output by the discriminator. The second output represents whether the first output corresponds to the style option.



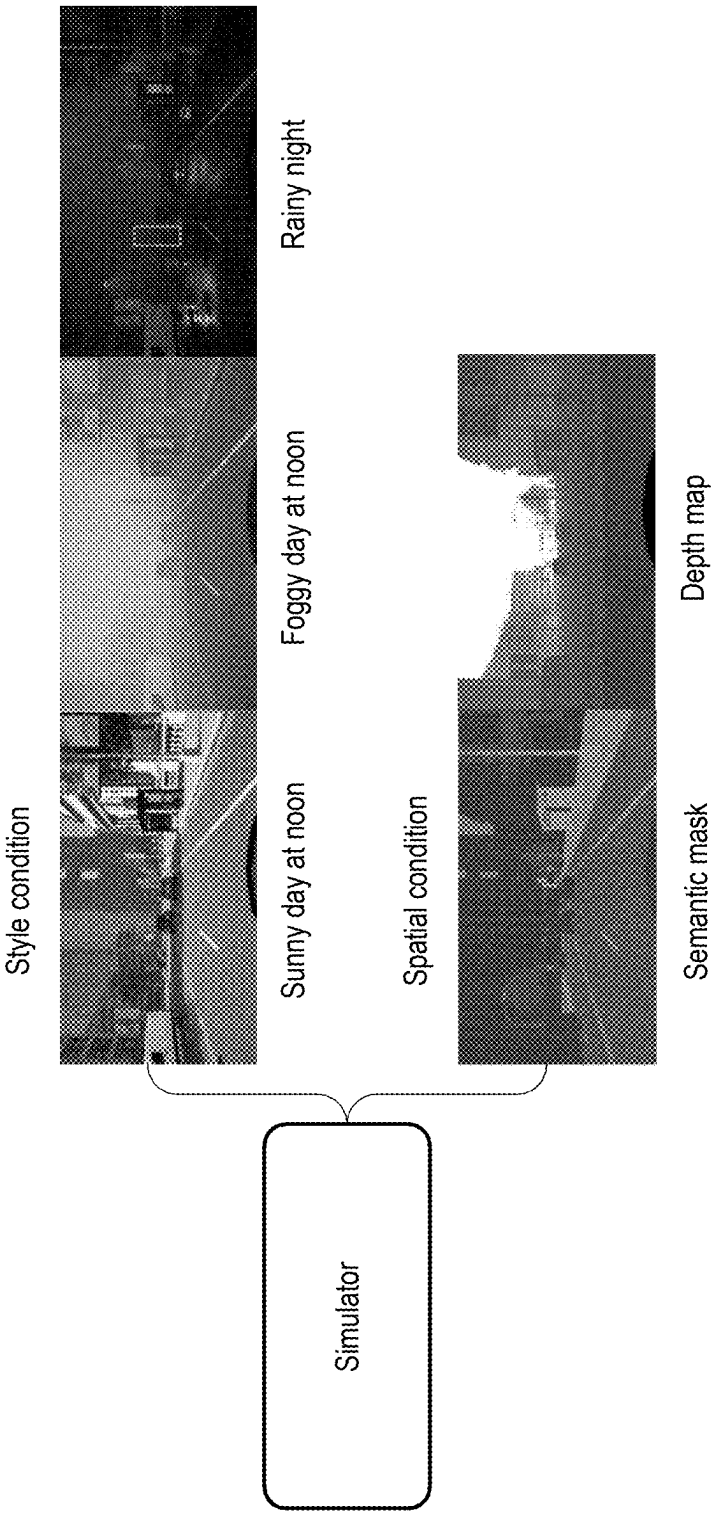


FIG. 1 (PRIOR ART)

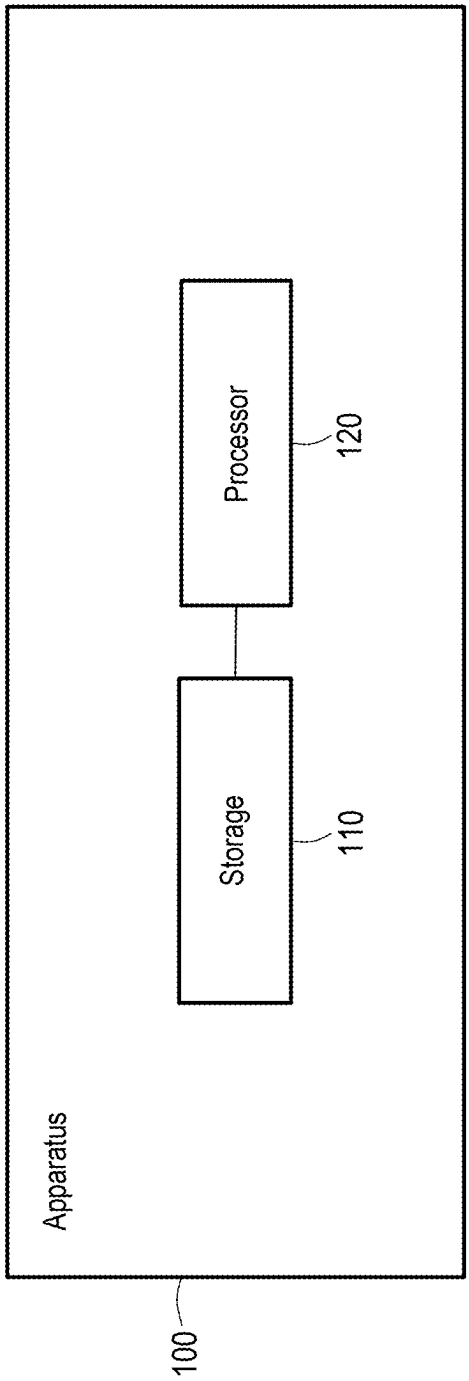


FIG. 2

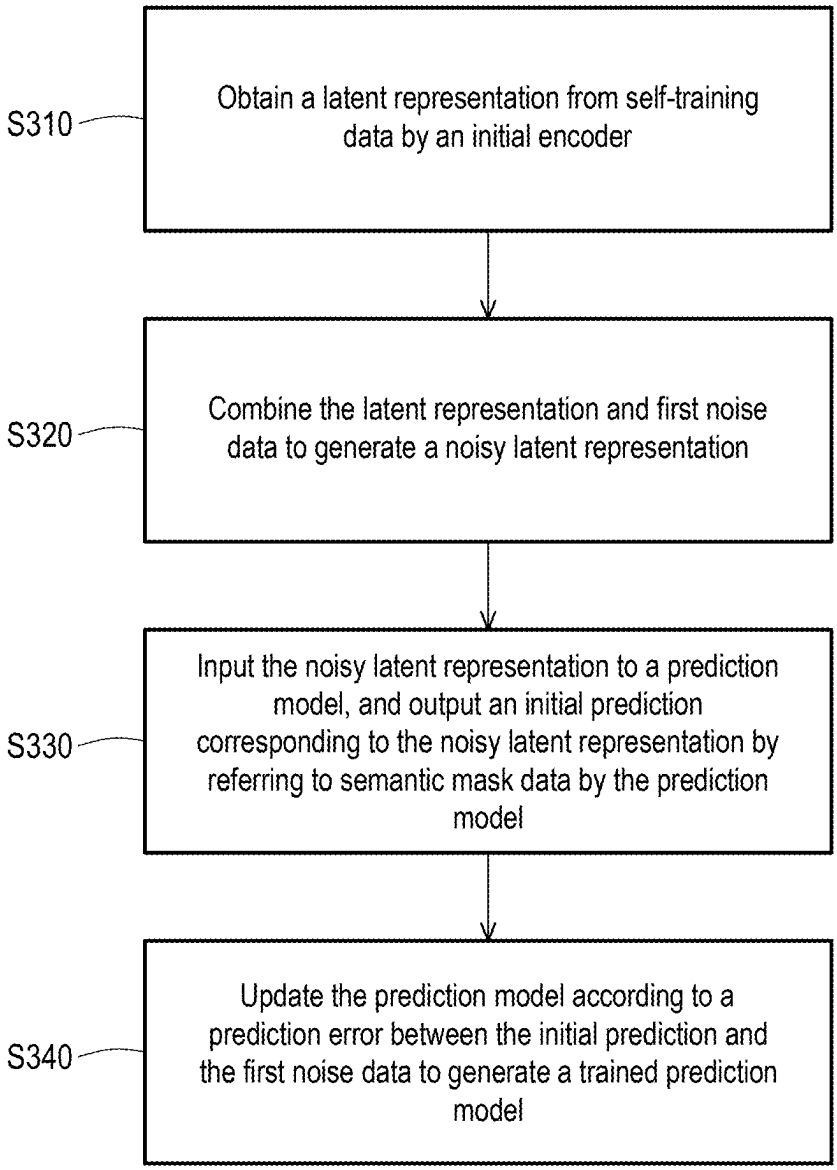

S310
Obtain a latent representation from self-training data by an initial encoder
S320
Combine the latent representation and first noise data to generate a noisy latent representation
S330
Input the noisy latent representation to a prediction model, and output an initial prediction corresponding to the noisy latent representation by referring to semantic mask data by the prediction model
S340
Update the prediction model according to a prediction error between the initial prediction and the first noise data to generate a trained prediction model

FIG. 3

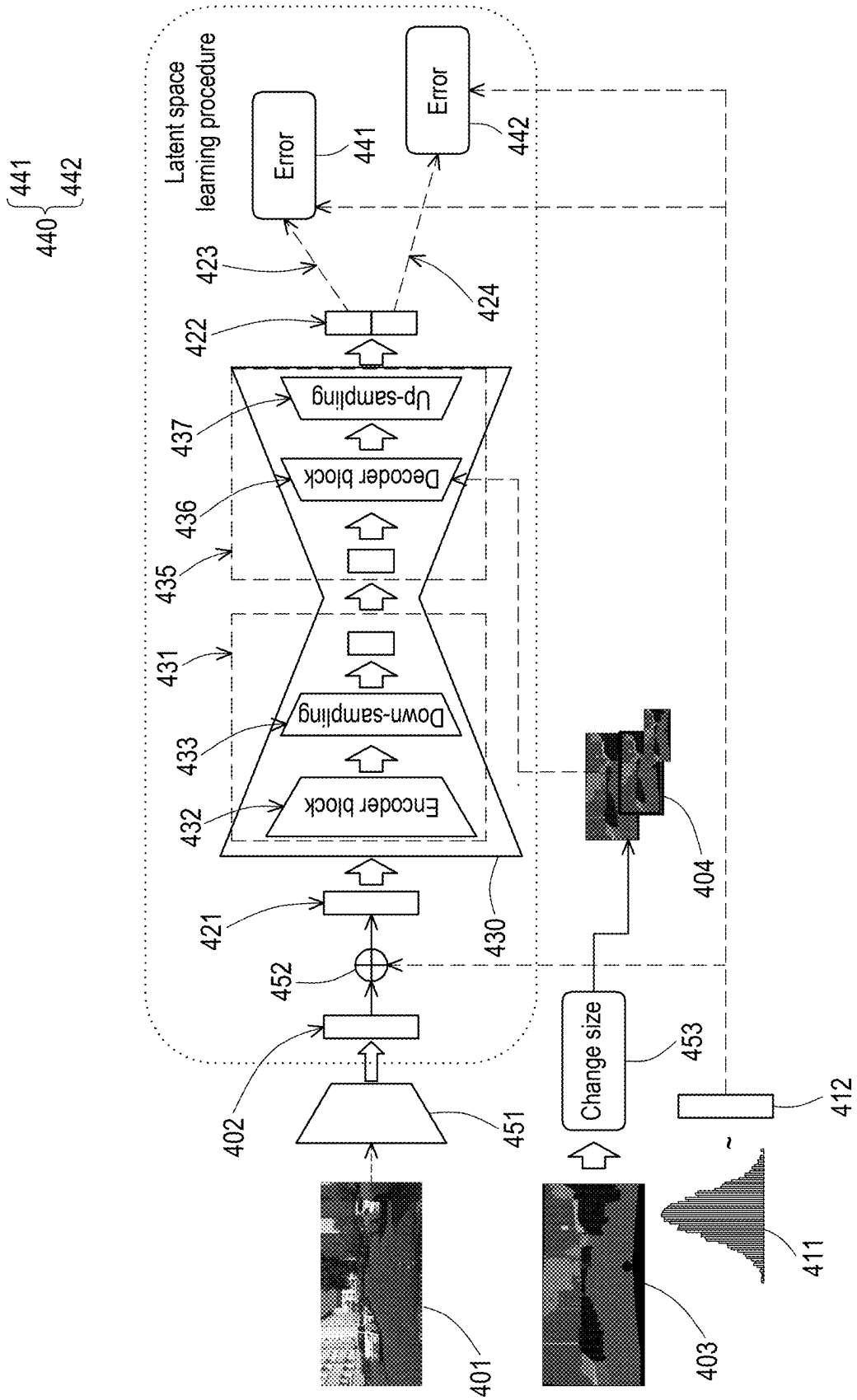

440
441
442
Latent space learning procedure
Error
441
Error
442
423
424
422
Up-sampling
437
Decoder block
436
435
431
Down-sampling
433
Encoder block
432
421
430
452
402
451
Change size
453
412
404
401
403
411

FIG. 4

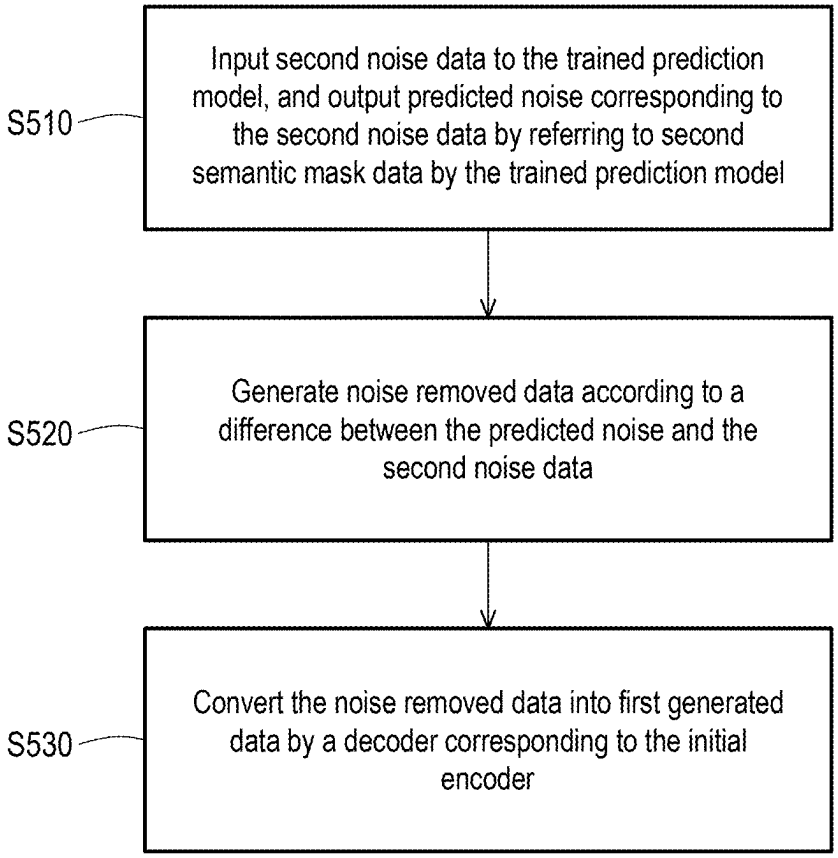

S510
Input second noise data to the trained prediction model, and output predicted noise corresponding to the second noise data by referring to second semantic mask data by the trained prediction model
S520
Generate noise removed data according to a difference between the predicted noise and the second noise data
S530
Convert the noise removed data into first generated data by a decoder corresponding to the initial encoder

FIG. 5

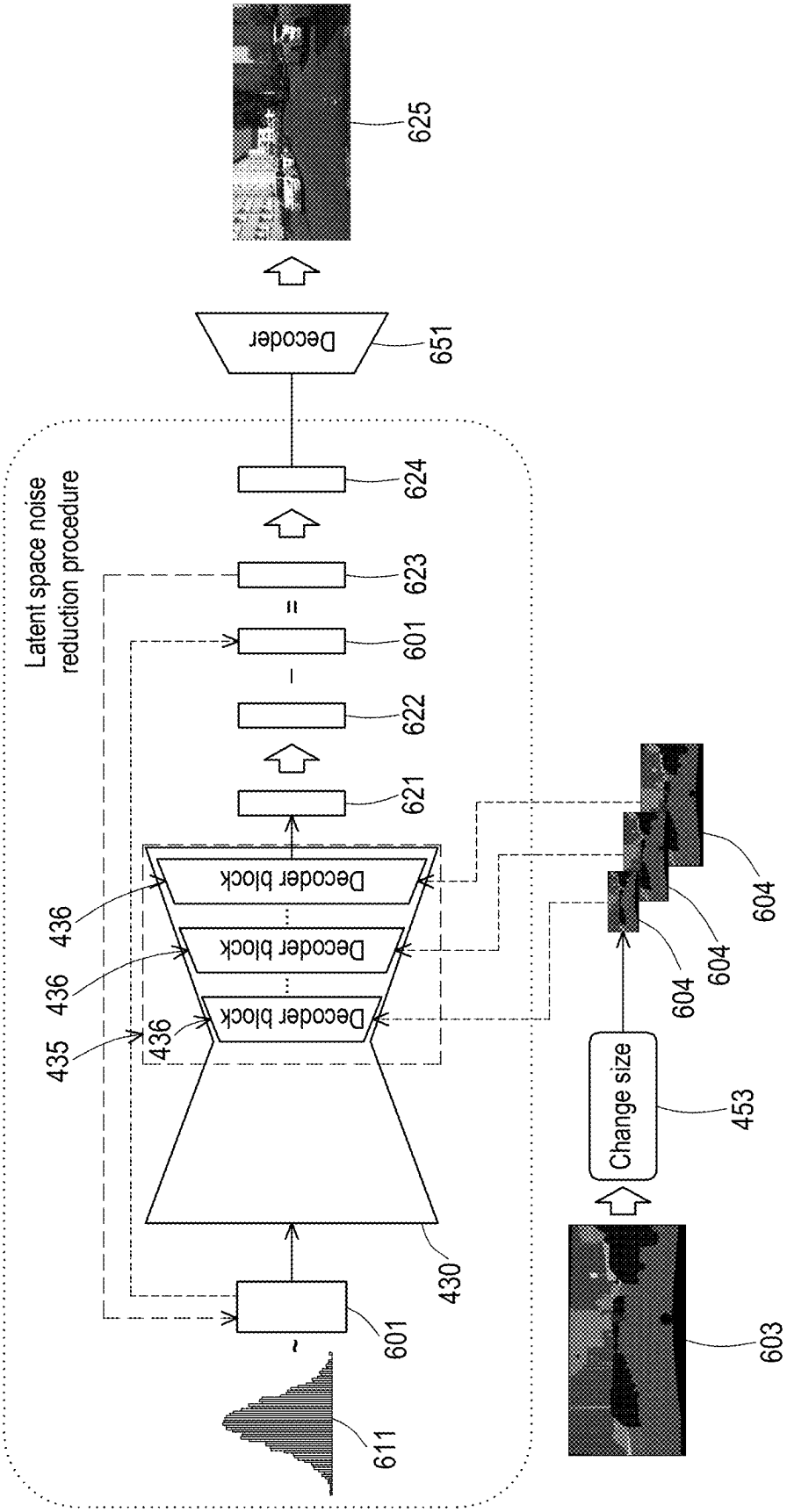

625
Decoder
651
624
623
601
622
621
Latent space noise reduction procedure
Decoder block
Decoder block
Decoder block
436
436
436
435
604
604
604
Change size
453
603
430
601
611

FIG. 6

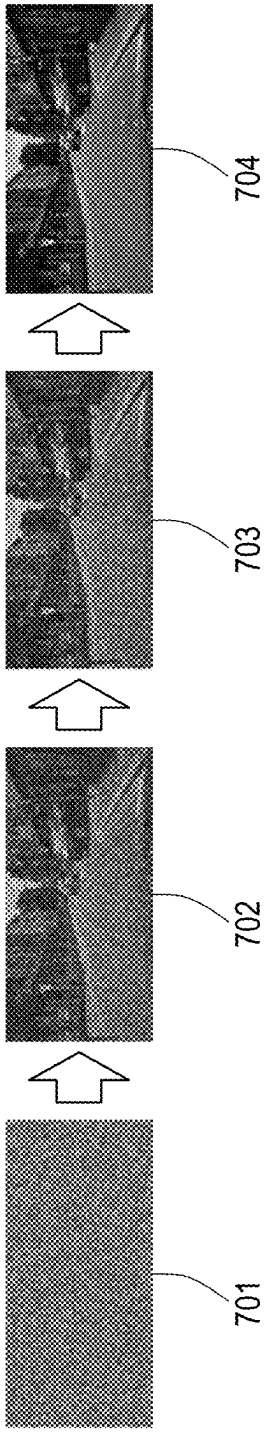


FIG. 7

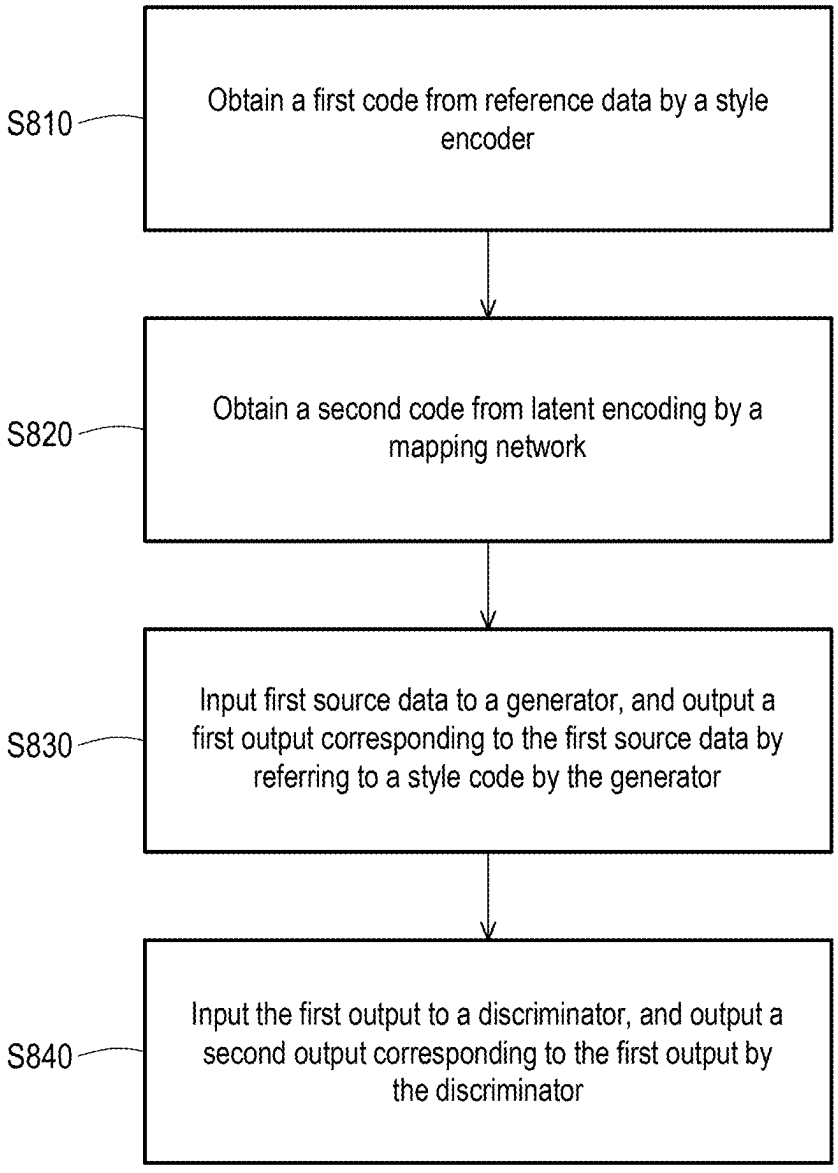

S810
Obtain a first code from reference data by a style encoder
S820
Obtain a second code from latent encoding by a mapping network
S830
Input first source data to a generator, and output a first output corresponding to the first source data by referring to a style code by the generator
S840
Input the first output to a discriminator, and output a second output corresponding to the first output by the discriminator

FIG. 8

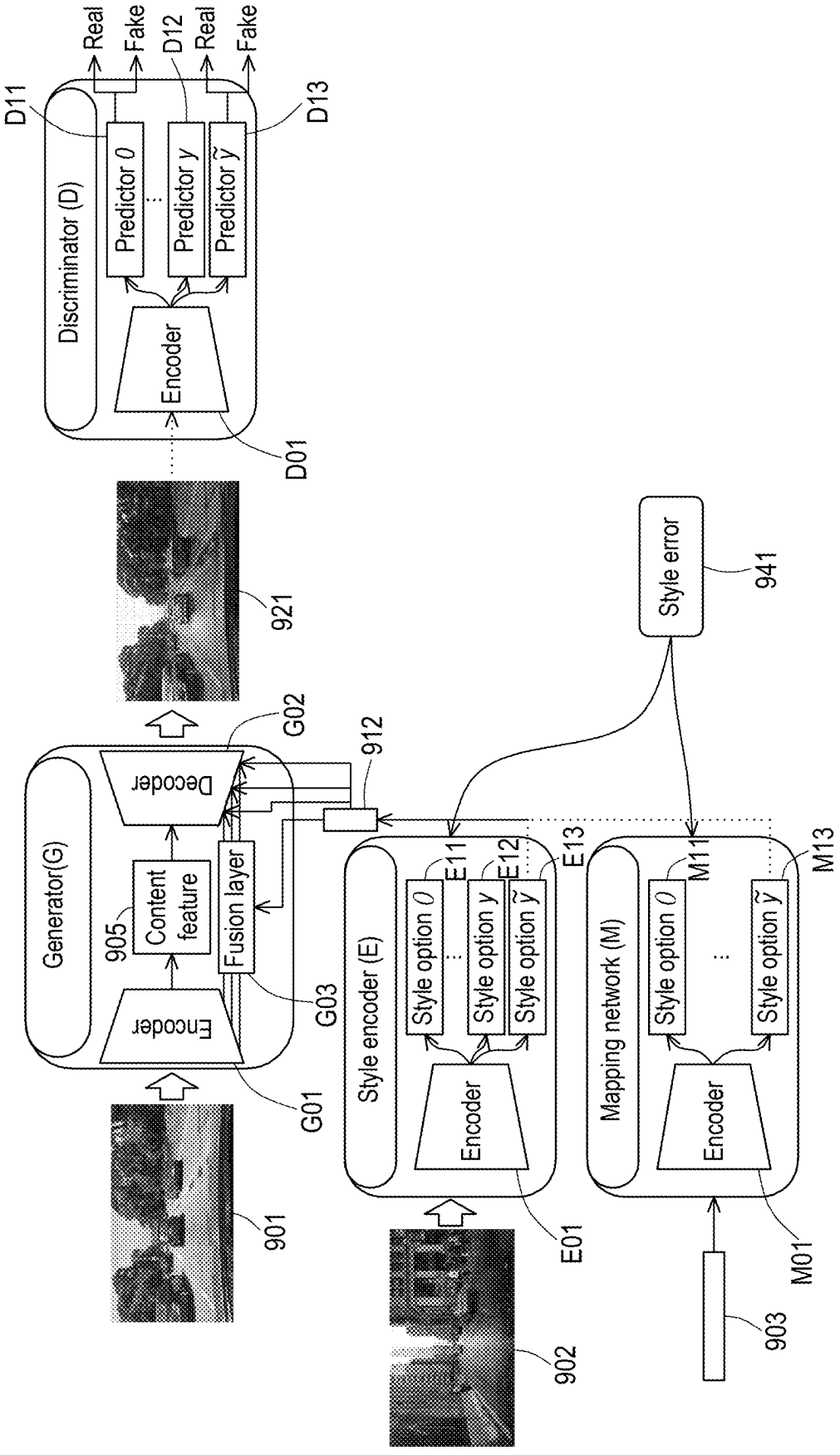

Real
Fake
D12
Real
Fake
D11
D13
Discriminator (D)
Predictor 0
Predictor y
Predictor ỹ
Encoder
D01
921
G02
912
Style error
941
Decoder
Generator(G)
Content feature
905
Fusion layer
Encoder
G03
G01
Style encoder (E)
E11
E12
E13
Style option 0
Style option y
Style option ỹ
Encoder
E01
Mapping network (M)
M11
M13
Style option 0
Style option ỹ
Encoder
M01
901
902
903

FIG. 9

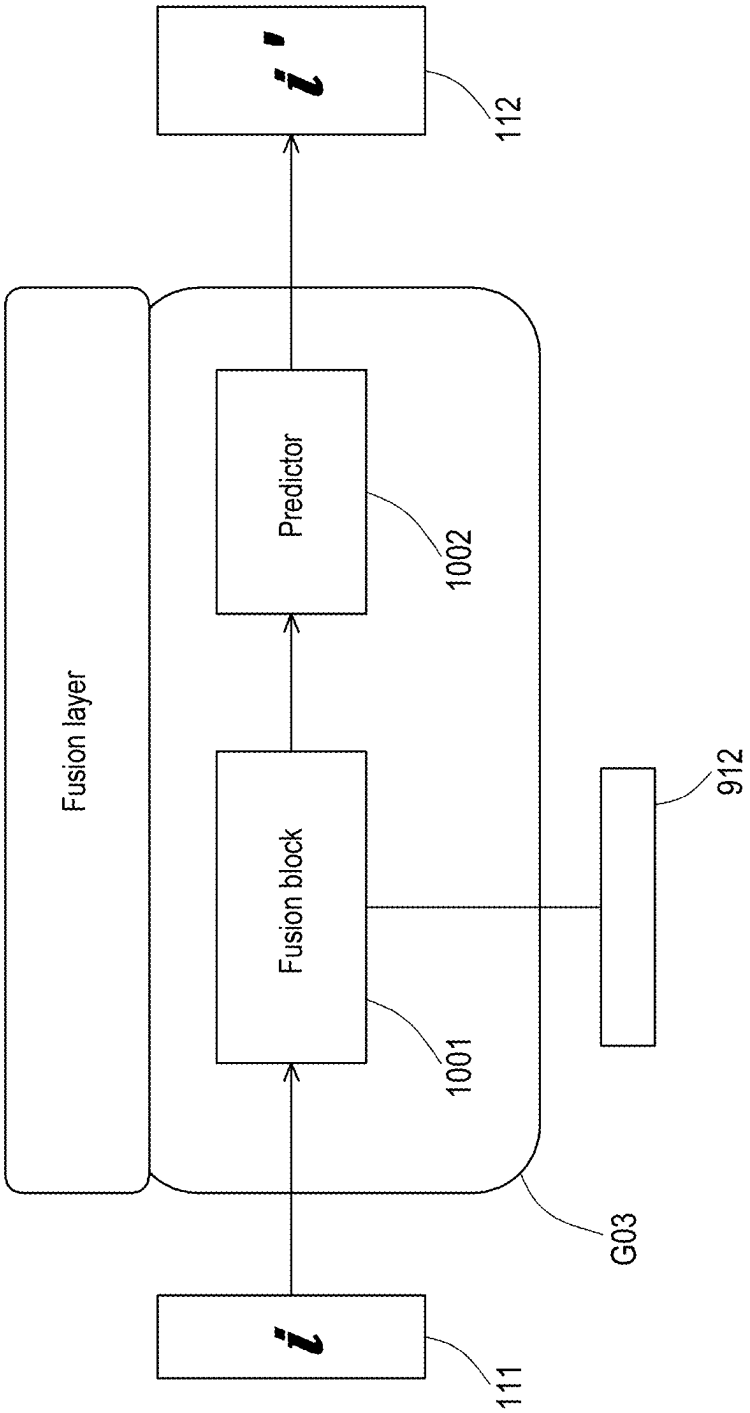

i'
112
Predictor
1002
Fusion layer
Fusion block
1001
912
G03
i
111

FIG. 10

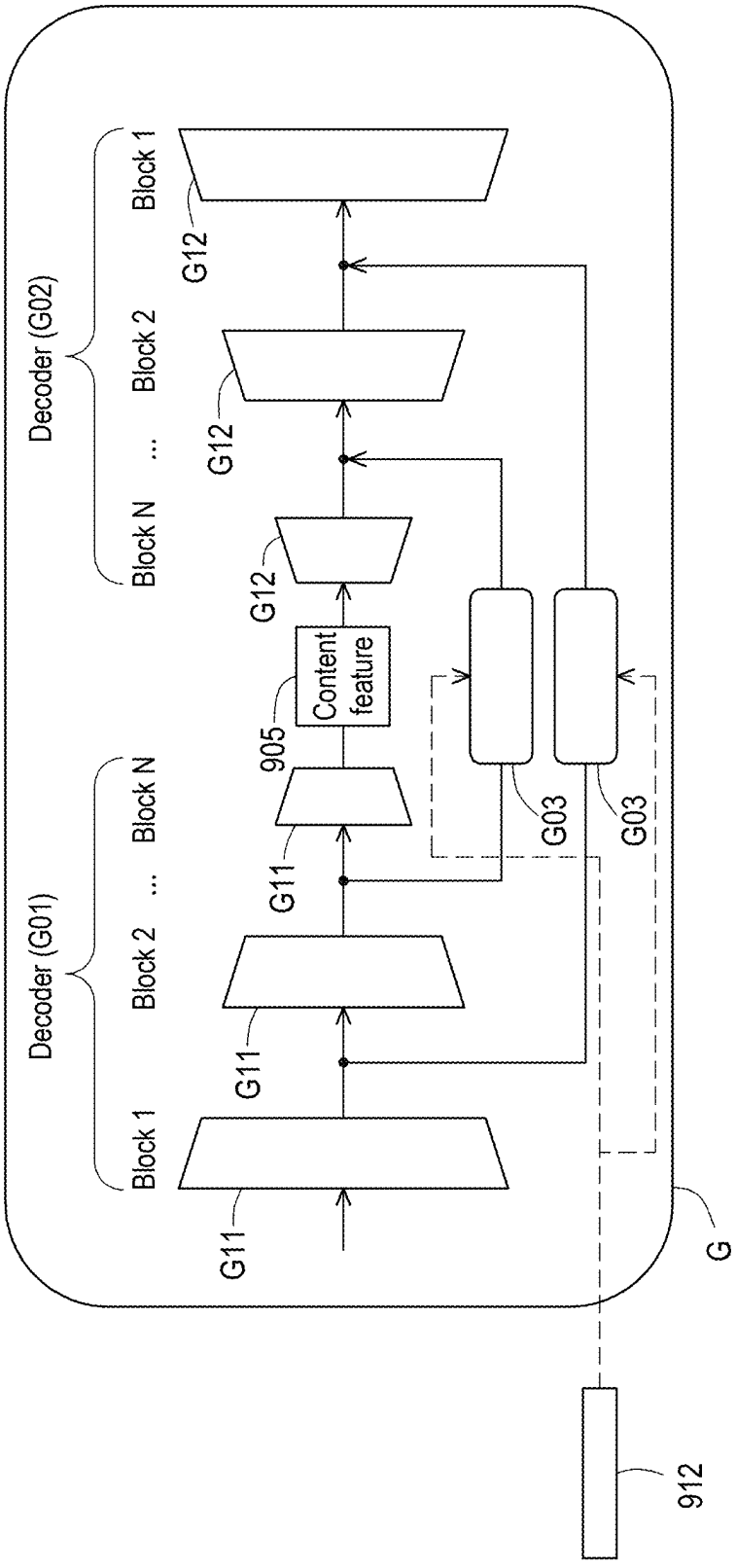

Decoder (G01)
Block 1
Block 2
...
Block N
Decoder (G02)
Block 1
Block 2
...
Block N
G11
G11
G11
G12
G12
G12
905
Content feature
G03
G03
G
912

FIG. 11

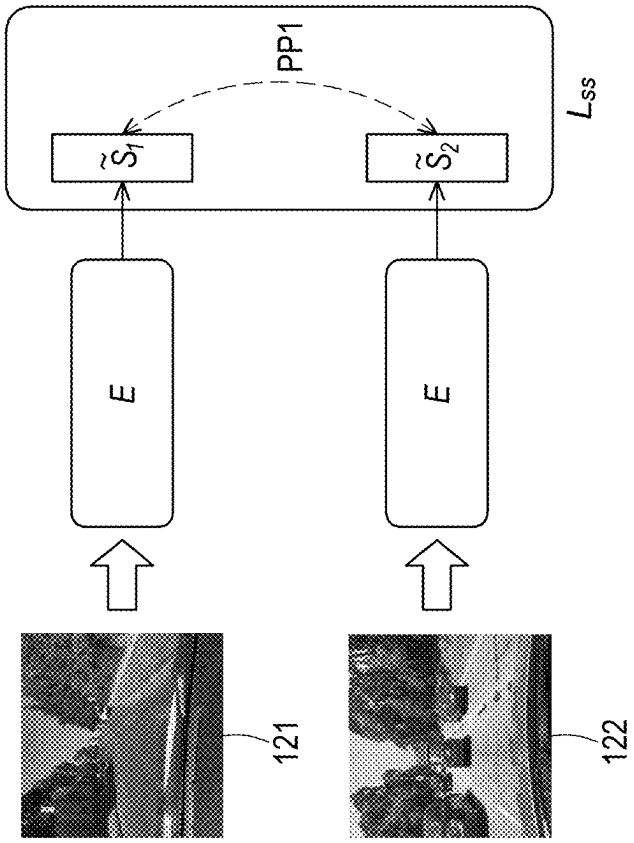


FIG. 12A

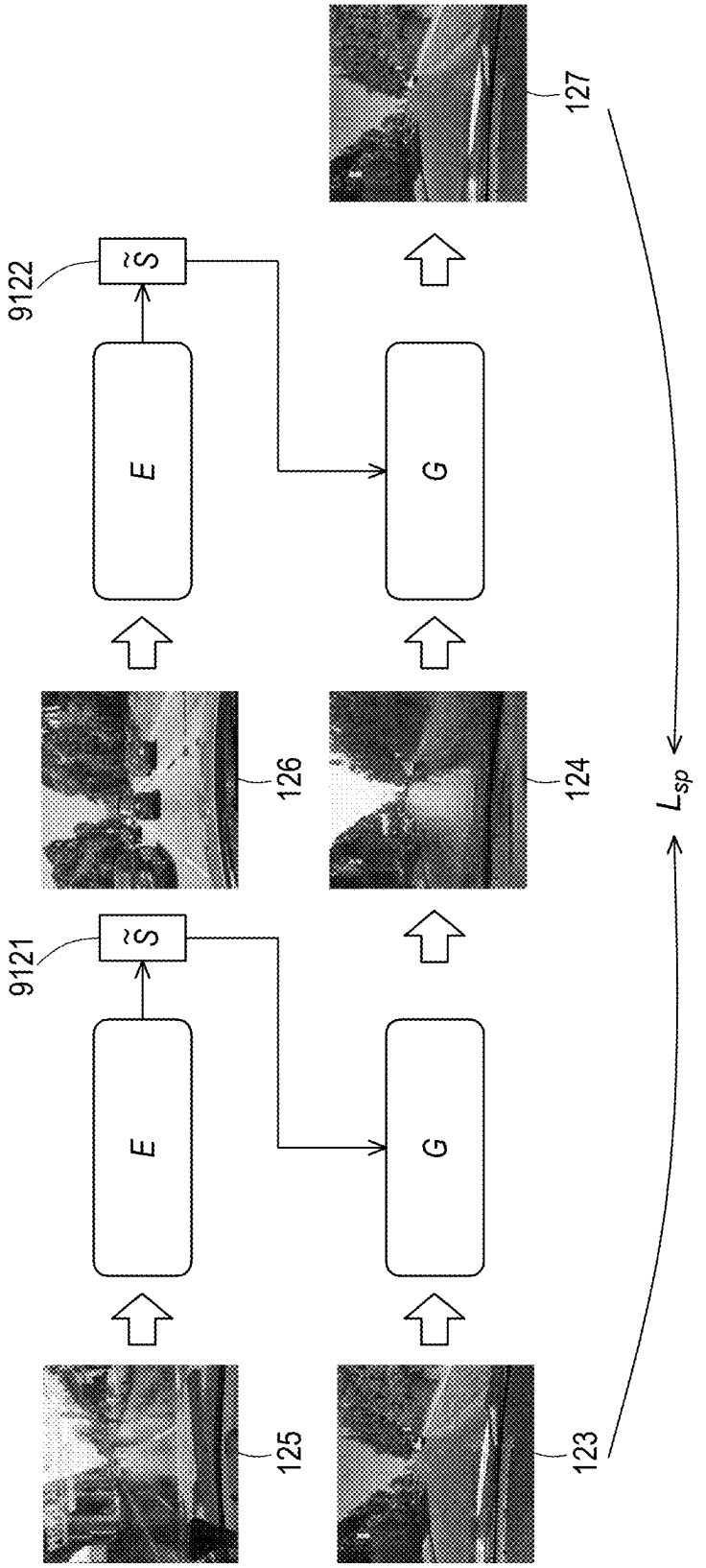


FIG. 12B

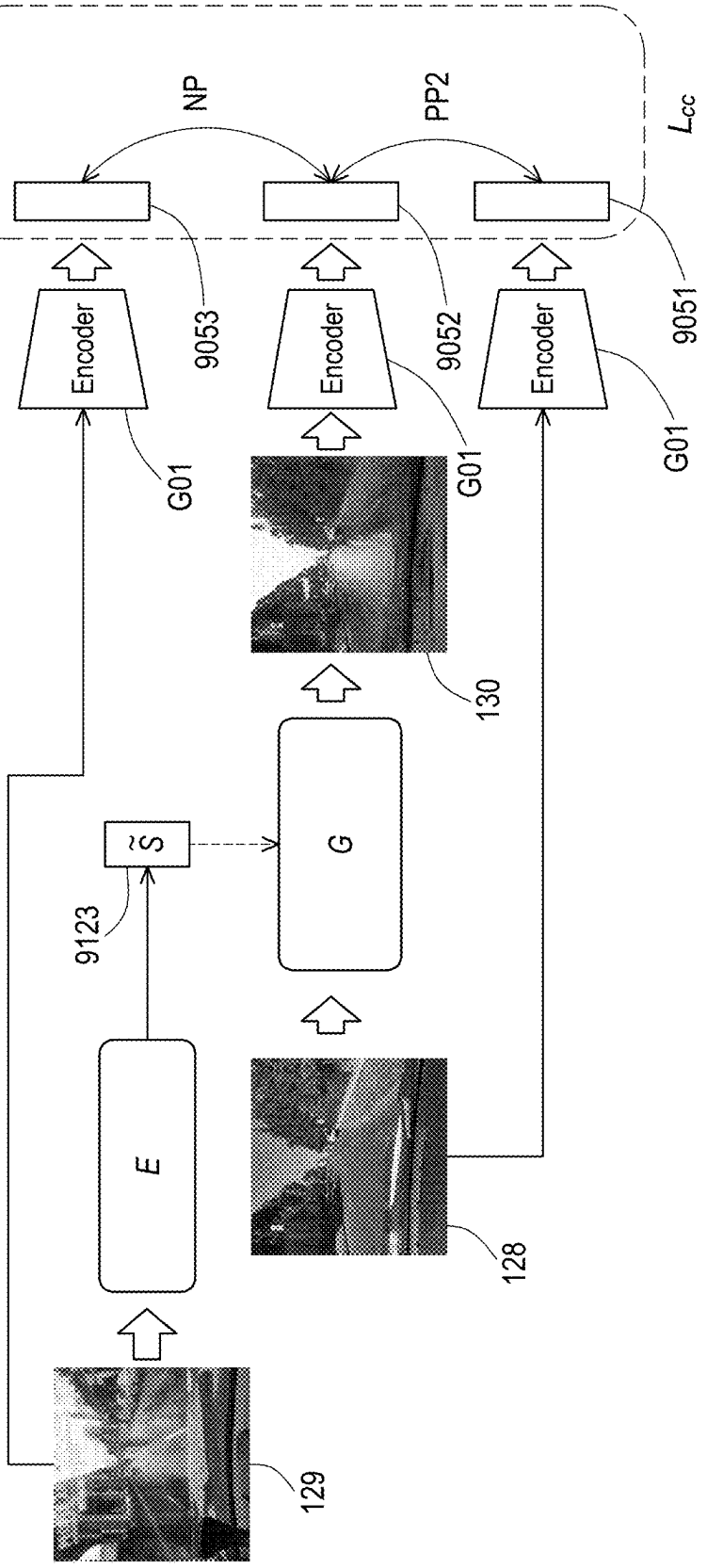

NP
PP2
L_{cc}
Encoder
Encoder
Encoder
9053
9052
9051
G01
G01
G01
130
G
$\tilde{S}$
9123
E
128
129

FIG. 12C

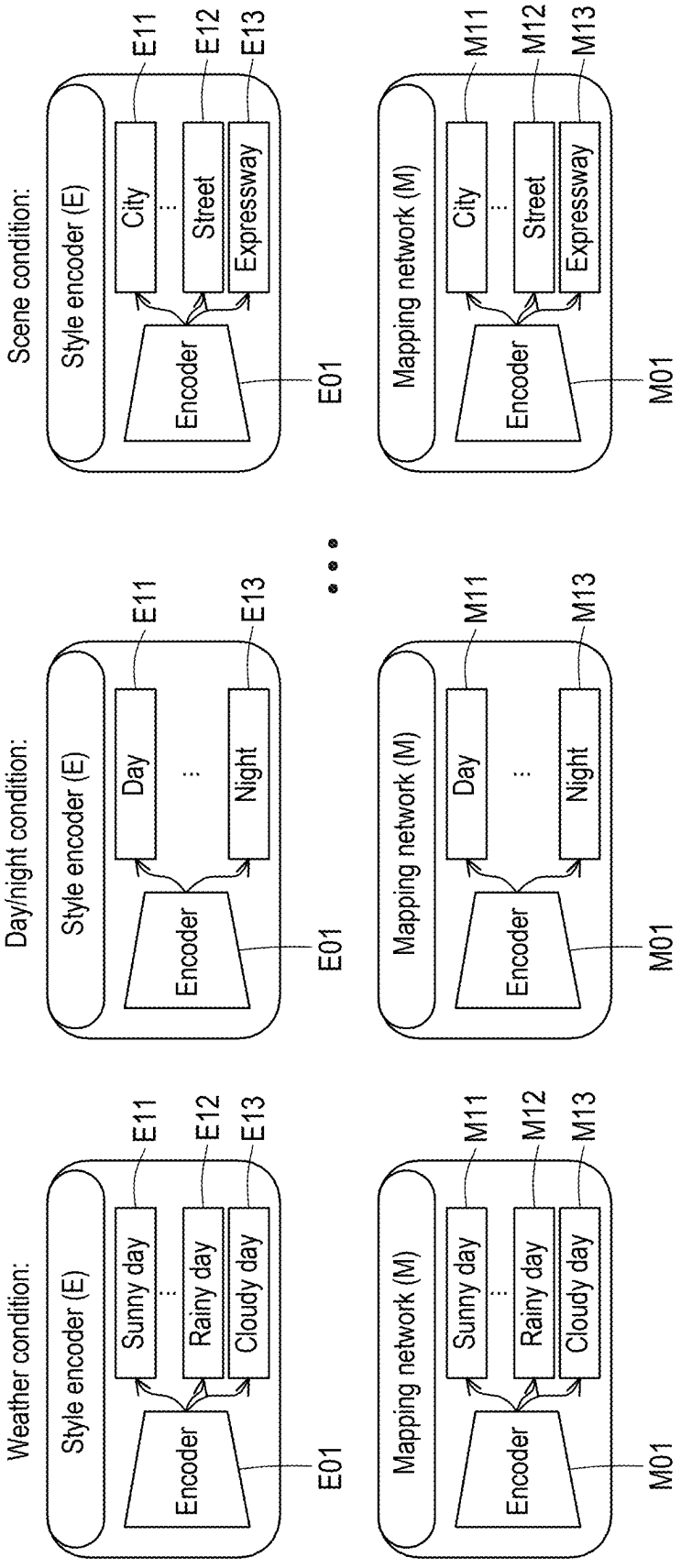

Scene condition:
Style encoder (E)
E11
E12
E13
City
Street
Expressway
Encoder
E01
Mapping network (M)
M11
M12
M13
M01
Day/night condition:
Day
Night
Weather condition:
Sunny day
Rainy day
Cloudy day

FIG. 13

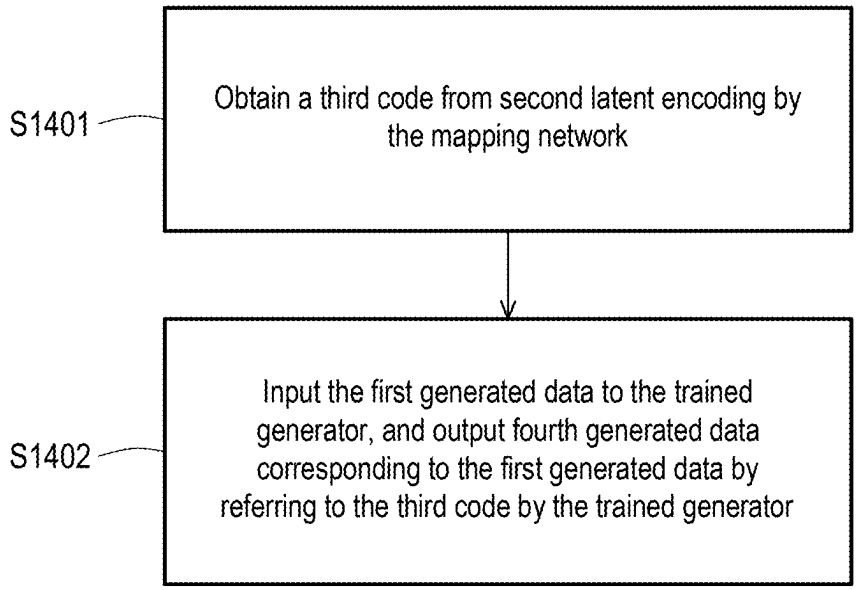

S1401
Obtain a third code from second latent encoding by the mapping network
S1402
Input the first generated data to the trained generator, and output fourth generated data corresponding to the first generated data by referring to the third code by the trained generator

FIG. 14

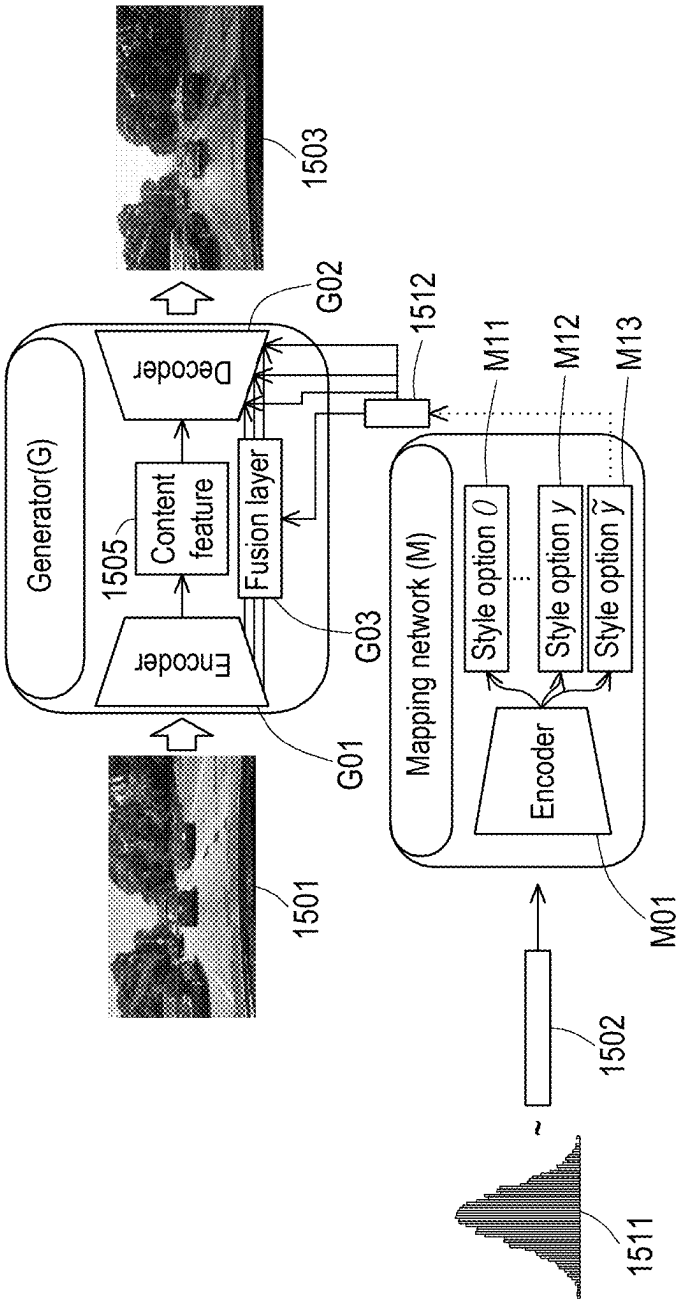

1503
G02
1512
M11
M12
M13
Decoder
Generator(G)
Content feature
1505
Fusion layer
Encoder
G03
G01
Mapping network (M)
Style option 0
Style option y
Style option ỹ
Encoder
1501
M01
1502
z
1511

FIG. 15

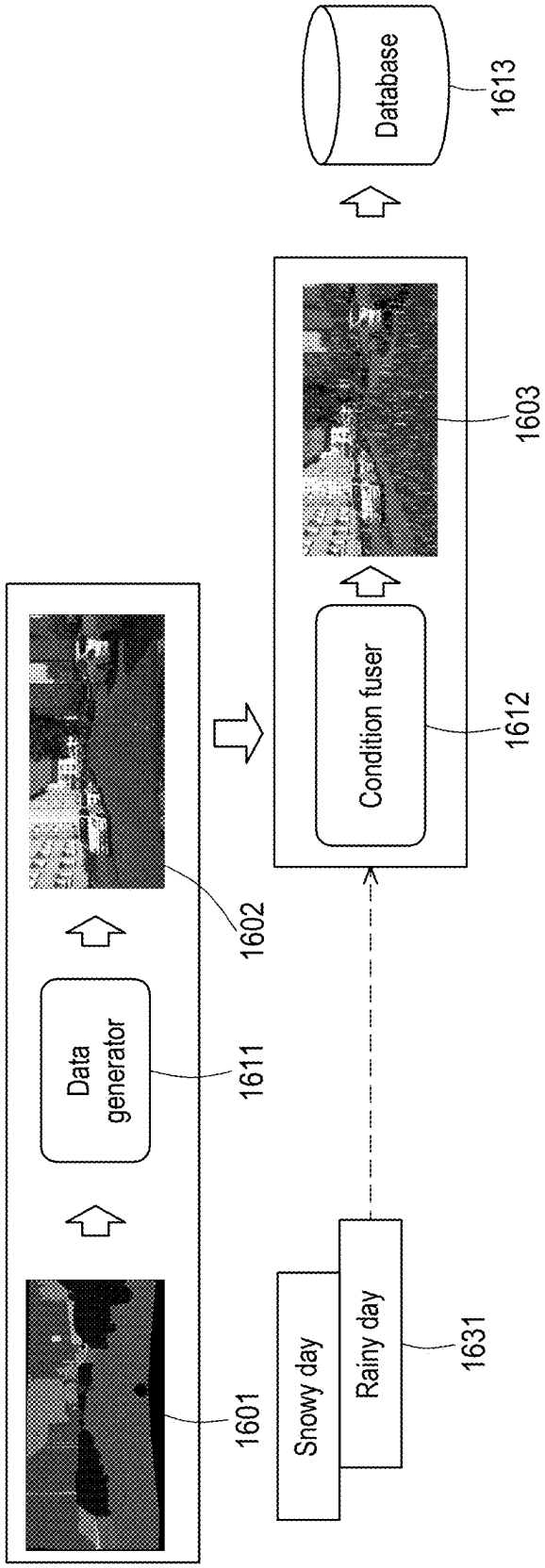

Database
1613
1603
Condition fuser
1612
1602
Data generator
1611
Snowy day
Rainy day
1631
1601

FIG. 16A

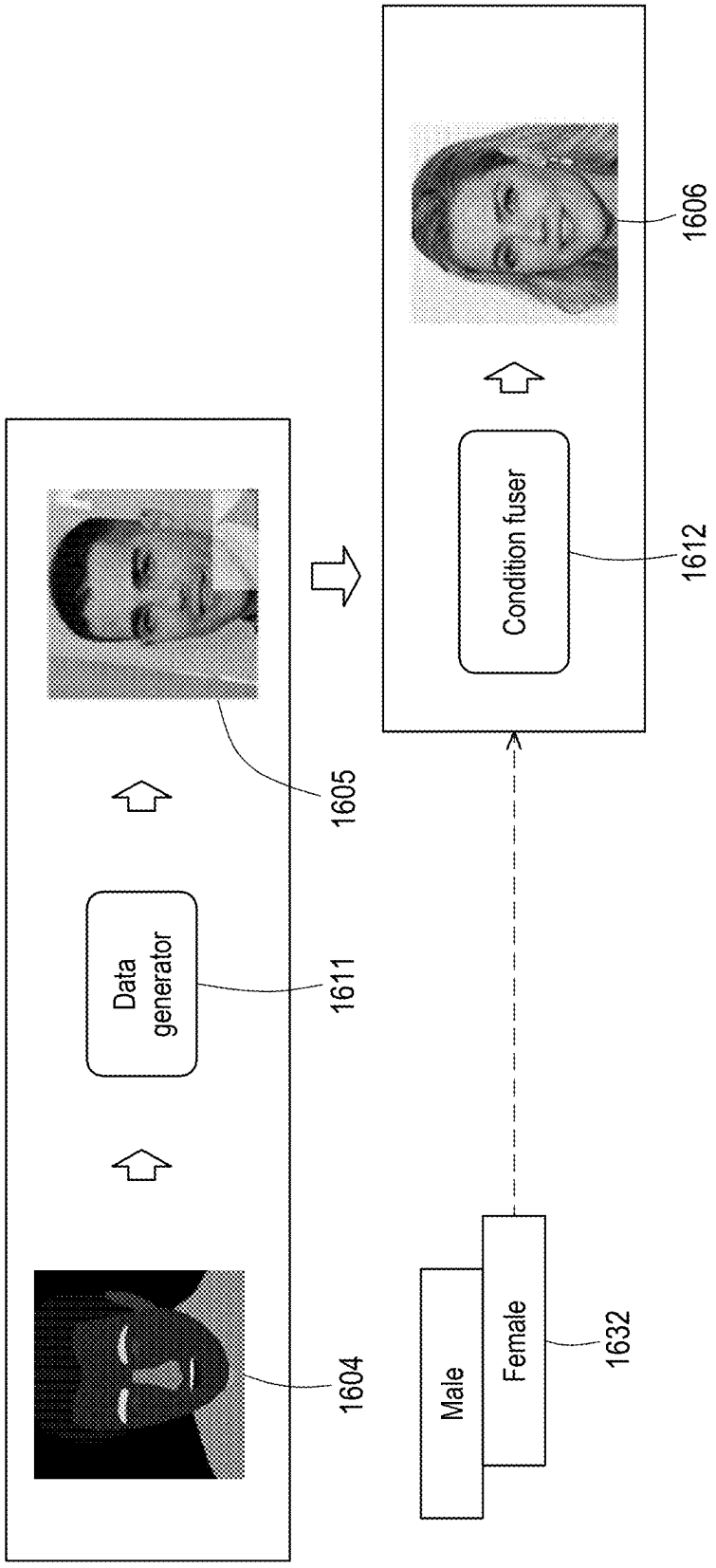

1606
Condition fuser
1612
1605
Data generator
1611
1604
Male
Female
1632

FIG. 16B

FIG. 17A

FIG. 17B

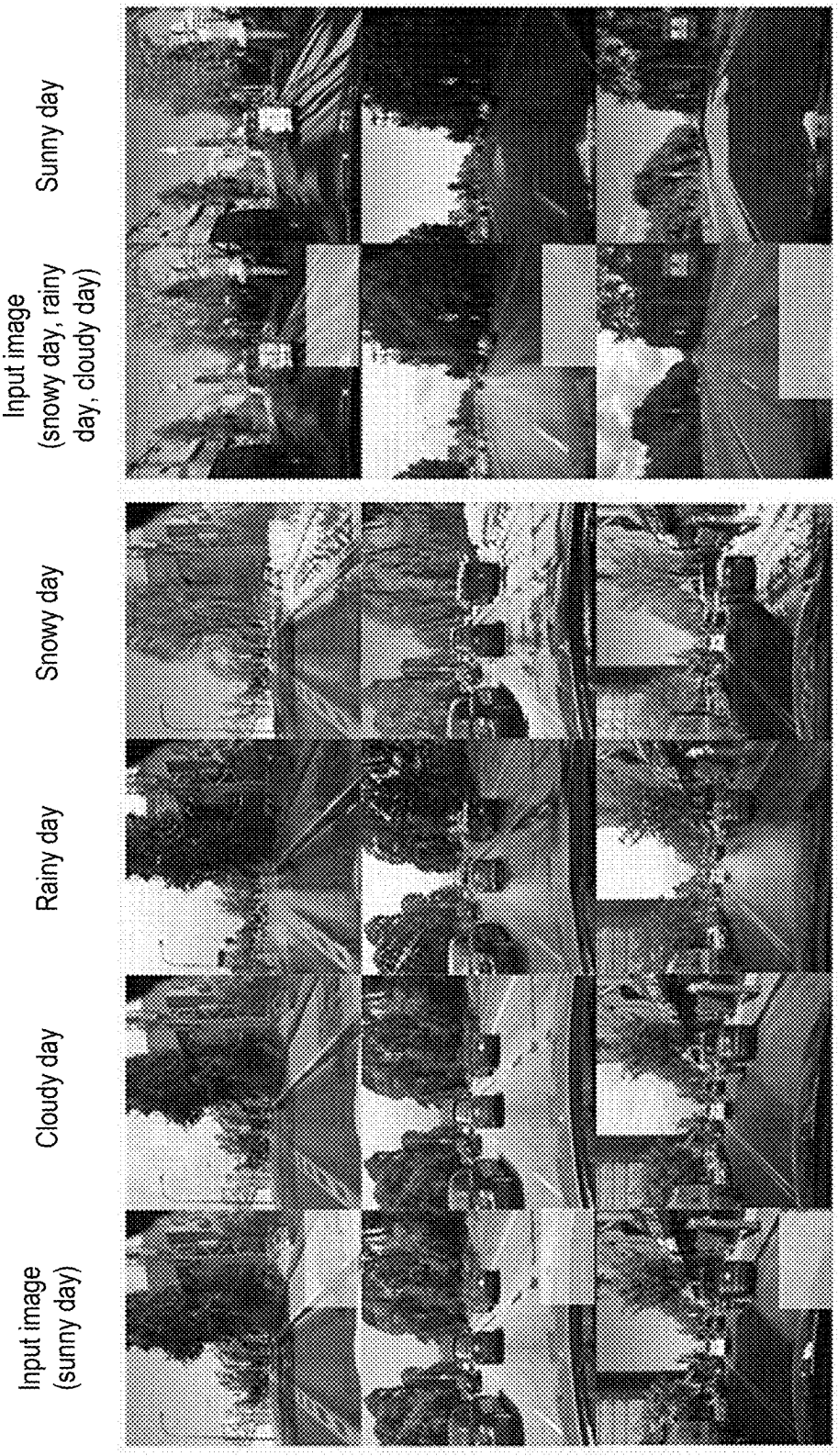

Input image (sunny day)
Cloudy day
Rainy day
Snowy day
Input image (snowy day, rainy day, cloudy day)
Sunny day

FIG. 18

METHOD AND APPARATUS RELATED TO DATA GENERATION FRAMEWORK

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of U.S. provisional application Ser. No. 63/560,790, filed on Mar. 4, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to a data generation technology, and in particular to a method and an apparatus related to a data generation framework.

Description of Related Art

[0003] FIG. **1** is a schematic diagram of a simulation scenario. Please refer to FIG. **1**. In the prior art, a simulator may generate corresponding images according to customized conditions. For example, the customized style conditions are sunny day at noon, foggy day at noon, and rainy night. Alternatively, the customized spatial conditions are semantic mask and depth map that define corresponding categories of image regions. However, the conventional simulator can only generate fake images in a virtual scenario, which is even difficult to be applied in designing various three-dimensional objects.

SUMMARY

[0004] The disclosure provides a method and an apparatus related to a data generation framework, which may generate more realistic data according to customized conditions.

[0005] A method related to a data generation framework according to an embodiment of the disclosure is implemented by a processor and includes the following steps of: obtaining a first code from reference data by a style encoder; obtaining a second code from latent encoding by a mapping network, wherein the first code and the second code are both a style code, and the style code corresponds to at least one style option; inputting first source data to a generator, and outputting a first output corresponding to the first source data by referring to the style code by the generator; and inputting the first output to a discriminator, and outputting a second output corresponding to the first output by the discriminator, wherein the second output represents whether the first output corresponds to the at least one style option.

[0006] An apparatus related to a data generation framework according to an embodiment of the disclosure includes (but is not limited to) a storage and a processor. The storage is used to store a program code. The processor is coupled to the storage. The processor is configured to load the program code to execute: obtaining a first code from reference data by a style encoder; obtaining a second code from latent encoding by a mapping network, wherein the first code and the second code are both a style code, and the style code corresponds to at least one style option; inputting first source data to a generator, and outputting a first output corresponding to the first source data by referring to the style code by the generator; and inputting the first output to a discriminator, and outputting a second output corresponding to the first output by the discriminator, wherein the second output represents whether the first output corresponds to the at least one style option.

[0007] Based on the above, the method and the apparatus related to the data generation framework according to the embodiments of the disclosure learn how to change the input data into real data using the style code through the generator, and judge whether the input data is the real data through the discriminator. In this way, high quality, realistic, and reliable data generation may be provided.

[0008] In order for the features and advantages of the disclosure to be more comprehensible, the following specific embodiments are described in detail in conjunction with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is a schematic diagram of a simulation scenario.

[0010] FIG. **2** is an element block diagram of an apparatus related to a data generation framework according to an embodiment of the disclosure.

[0011] FIG. **3** is a flowchart of spatial learning related to a data generation framework according to an embodiment of the disclosure.

[0012] FIG. **4** is a schematic diagram of spatial learning related to a data generation framework according to an embodiment of the disclosure.

[0013] FIG. **5** is a flowchart of spatial noise removal related to a data generation framework according to an embodiment of the disclosure.

[0014] FIG. **6** is a schematic diagram of spatial noise removal related to a data generation framework according to an embodiment of the disclosure.

[0015] FIG. **7** is a schematic diagram illustrating noise removal according to an embodiment of the disclosure.

[0016] FIG. **8** is a flowchart of multi-domain conditioned learning related to a data generation framework according to an embodiment of the disclosure.

[0017] FIG. **9** is a schematic diagram of multi-domain conditioned learning related to a data generation framework according to an embodiment of the disclosure.

[0018] FIG. **10** is a schematic diagram of a fusion layer according to an embodiment of the disclosure.

[0019] FIG. **11** is a schematic diagram of a generator according to an embodiment of the disclosure.

[0020] FIG. **12**A is a schematic diagram of a style similarity error according to an embodiment of the disclosure.

[0021] FIG. **12**B is a schematic diagram of a source preservation error according to an embodiment of the disclosure.

[0022] FIG. **12**C is a schematic diagram of a content comparison error according to an embodiment of the disclosure.

[0023] FIG. **13** is a schematic diagram of a style option according to an embodiment of the disclosure.

[0024] FIG. **14** is a flowchart of inference of multi-domain conditions related to a data generation framework according to an embodiment of the disclosure.

[0025] FIG. **15** is a schematic diagram of inference of multi-domain conditions related to a data generation framework according to an embodiment of the disclosure.

[0026] FIG. **16**A is a schematic diagram of scene image generation according to an embodiment of the disclosure.

[0027] FIG. **16**B is a schematic diagram of facial image generation according to an embodiment of the disclosure.

[0028] FIG. **17**A and FIG. **17**B are experimental results of a spatial conditioned pipeline according to an embodiment of the disclosure.

[0029] FIG. **18** is an experimental result of a multi-domain conditioned pipeline according to an embodiment of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

[0030] FIG. **2** is an element block diagram of an apparatus **100** related to a data generation framework according to an embodiment of the disclosure. Please refer to FIG. **2**. The apparatus **100** includes (but is not limited to) a storage **110** and a processor **120**. The apparatus **100** may be a mobile phone, a tablet computer, a notebook computer, a desktop computer, a server, a voice assistant apparatus, a smart home appliance, a wearable apparatus, a vehicle system, or other electronic apparatuses.

[0031] The storage **110** may be any type of fixed or removable random access memory (RAM), read only memory (ROM), flash memory, hard disk drive (HDD), solid state drive (SSD), or similar elements. In an embodiment, the storage **110** is used to store program codes, software modules, configurations, data (for example, model parameters, data sets, samples, features, or predictions), or files, which will be described in detail in subsequent embodiments.

[0032] The processor **120** is coupled to the storage **110**. The processor **120** may be a central processing unit (CPU), a graphics processing unit (GPU), other programmable general-purpose or specific-purpose microprocessors, digital signal processors (DSP), programmable controllers, field programmable gate arrays (FPGA), application-specific integrated circuits (ASIC), neural processing units (NPU), tensor processing units (TPU), artificial intelligence (AI) accelerators, neural engines, other similar elements, or a combination of the above elements. In an embodiment, the processor **120** is used to execute all or some operations of the apparatus **100** and may load and execute various program codes, software modules, files, and data stored in the storage **110**.

[0033] In the following, a method described in an embodiment of the disclosure will be illustrated with reference to various apparatuses, elements, and modules in the apparatus **100**. Each procedure of the method may be adjusted according to the implementation situation and is not limited thereto.

[0034] FIG. **3** is a flowchart of spatial learning related to a data generation framework according to an embodiment of the disclosure. Please refer to FIG. **3**. The processor **120** obtains a latent representation from self-training data by an initial encoder (step S**310**). Specifically, the training data may be images, such as image samples obtained from (training) data sets from various sources. For example, FIG. **4** is a schematic diagram of spatial learning related to a data generation framework according to an embodiment of the disclosure. Please refer to FIG. **4**. Training data **401** is an image of a street. In other embodiments, the training data may be sound, text, sensing intensity, angle, amplitude, position, or other forms of data.

[0035] In an embodiment, an initial encoder **451** is a vector quantized (VQ) encoder. The initial encoder **451** may divide a continuous high-dimensional vector space into several regions, and each region corresponds to one representative performance/representation (for example, a (compressed) vector, a code vector, or a codeword form). During an encoding process of the initial encoder **451**, the processor **120** maps the input training data to a closest latent representation **402**. The initial encoder **451** may map high-dimensional data to a low-dimensional space, while retaining important information of the data. Such low-dimensional performance/representation is often referred to as the latent representation **402**.

[0036] In another embodiment, the initial encoder **451** is another encoder for dimensionality reduction or compression.

[0037] Please refer to FIG. **3**. The processor **120** combines the latent representation and first noise data to generate a noisy latent representation (step S**320**). Specifically, the first noise data may be values based on a statistical distribution such as a Gaussian distribution, a uniform distribution, or a Poisson distribution or data generated based on noise in a real environment. It should be noted that to maximize the diversity of generated data, most techniques directly employ the Gaussian distribution, which has the maximum entropy. Other candidate distributions are not as effective in generating diversity, but they may be selected based on specific circumstances. Taking FIG. **4** as an example, the processor **120** samples a Gaussian distribution **411**, and generates first noise data **412** according to sampled values. Next, the processor **120** adds the latent representation **402** and the first noise data **412**, and uses the addition result as a noisy latent representation **421**.

[0038] In an embodiment, the latent representation and the first noise data are in the vector form. The processor **120** may add the latent representation and the first noise data element-wise, and use the addition result as the noisy latent representation. For example, a first element of the latent representation is added to a first element of the first noise data to become a first element of the noisy latent representation, a second element of the latent representation is added to a second element of the first noise data to become a second element of the noisy latent representation, and the rest may be deduced by analogy and will not be repeated here.

[0039] Please refer to FIG. **3**. The processor **120** inputs the noisy latent representation to a prediction model, and outputs an initial prediction corresponding to the noisy latent representation by referring to semantic mask data by the prediction model (step S**330**). Specifically, in a training phase, the noisy latent representation and a ground truth (for example, the first noise data) are used as training samples of the prediction model. The processor **120** may train the prediction model through a machine learning algorithm. The machine learning algorithm may analyze marked training samples (for example, the noisy latent representation with the corresponding ground truth) to establish a correlation between the latent representation embedded/added with noise (that is, the input of the model, such as the noisy latent representation **421** shown in FIG. **4**) and the embedded/added noise (that is, the output of the model, such as an initial prediction **422** shown in FIG. **4**). The first noise data **412** would be considered as the ground truth. The output of the model should be close to the ground truth. That is the model should be trained so that its output approximates the first noise data **412**, such as Gaussian noise, hidden in the input (i.e., the noisy latent representation **421**). The predic-

tion model may be learned, and inference on data to be evaluated (for example, a latent representation to be evaluated or other data) may be made accordingly to output an initial prediction (for example, noise data in the latent representation or other data) corresponding to the data to be evaluated.

[0040] The type of the machine learning algorithm may change according to the application scenario. The machine learning algorithm may be semantic latent diffusion, latent diffusion, or stochastic diffusion, but not limited thereto.

[0041] A prediction model **430** of FIG. **4** takes a semantic latent diffusion network as an example. The prediction model **430** includes an encoder **431** and a decoder **435**. The encoder **431** is connected to the decoder **435**. For example, an output (for example, a feature map) of the encoder **431** is used as an input of the decoder **435**. In an embodiment, for example, the encoder **431** and the decoder **435** have U-Net connection architectures.

[0042] The encoder **431** includes one or more encoder blocks **432**. The encoder block **432** is, for example, a diffusion encoder residual block (DER). A semantic latent diffusion residual block is a core element of an encoder part of a semantic latent diffusion model, and the main function thereof is to perform feature encoding or feature retrieval on input data with noise (for example, the noisy latent representation **421**). The encoder block **432** includes, for example, a convolutional layer, an activation function, a residual connection, and normalization processing, but not limited thereto.

[0043] In an embodiment, the encoder **431** includes multiple encoder blocks **432**. The processor **120** may perform down-sampling **433** on the output of the encoder block **432** to obtain data with smaller size or lower resolution. That is, the down-sampling **433** is used to reduce size or resolution. The data with smaller size or lower resolution may be input to the next encoder block **432**. In other words, the encoder blocks **432** respectively correspond to data with different sizes or resolutions.

[0044] The decoder **435** includes one or more decoder blocks **436**. The decoder block **436** is, for example, a diffusion decoder residual block (DDR). The semantic latent diffusion residual block is the core element of the encoder part of the semantic latent diffusion model, and the main function thereof is to combine features (also referred to as a feature map) retrieved by the encoder **431** and semantic information or spatial information (for example, first semantic mask data **403** and **404**, which will be introduced later). The decoder block **436** includes, for example, a deconvolutional layer, a convolutional layer, an activation function, a residual connection, and normalization processing, but not limited thereto.

[0045] In an embodiment, the decoder **435** includes multiple decoder blocks **436**. The processor **120** may perform up-sampling **437** on the output of the decoder block **436** to obtain data with larger size or higher resolution. That is, the up-sampling **437** is used to increase size or resolution. The data with larger size or higher resolution may be input to the next decoder block **436** or used as the output (that is, the initial prediction **422**) of the decoder **435** or the prediction model **430**. In other words, the decoder blocks **436** respectively correspond to data with different sizes or resolutions.

[0046] The first semantic mask data **403** and **404** define one or more first semantic categories for the training data **401**. The first semantic mask data **403** is marked data with the same size or resolution as the training data **401**. The first semantic mask data **403** and **404** are composed of multiple blocks, elements, or pixels. Taking the image form as an example, each pixel in the first semantic mask data **403** and **404** is assigned a predefined (first) semantic category (also referred to as label or semantic information). Taking a scene application as an example, the semantic category may be a lane, a car, a pedestrian, a building, or the sky.

[0047] In an embodiment, the processor **120** may generate the first semantic mask data **403** through a direct corresponding generator. The generator may generate the first semantic mask data **403** randomly or based on rules. In another embodiment, the processor **120** may receive a user operation, and define one or more first semantic categories in the first semantic mask data **403** and the first semantic category corresponding to the blocks, the elements, or the pixels in the first semantic mask data **403** according to parameters corresponding to the user operation.

[0048] In an embodiment, the processor **120** may generate the first semantic mask data **404** with multiple sizes (or multiple resolutions) (that is, change a size **453** or the resolution of the first semantic mask data **403**). The size or the resolution of the first semantic mask data **404** is smaller or lower than the size or the resolution of one or more first semantic mask data **403**. The first semantic mask data **404** with a certain size is aligned to the size of feature data (for example, the up-sampled feature map of the output of the encoder **431** or the output of another decoder block **436**) input to the corresponding decoder block **436**, and the first semantic mask data **404** and the feature data with the same size are input to the decoder block **436**. In an embodiment, in response to the decoder **435** including multiple decoder blocks **436**, the first semantic mask data **404** with multiple sizes or resolutions are respectively input to the same or corresponding decoder block **436** to increase sensitivity of the prediction model **430** to the semantic information with multiple resolutions or sizes.

[0049] In an embodiment, the decoder **435** or the decoder block **436** may adaptively adjust the mean and the variation (corresponding to a variation range of the mean) of the input feature map according to the first semantic mask data **403**.

[0050] In an embodiment, the initial prediction **422** includes a predicted mean **423** and a predicted variation **424** corresponding to the predicted mean **423**. The initial prediction **422** or the predicted mean **423** is, for example, the noise data predicted by the prediction model **430** for the noisy latent representation **421** (for example, the noise data predicted to be added or embedded to the noisy latent representation **421** or predicted first noise data).

[0051] Please refer to FIG. **3**. The processor **120** updates the prediction model according to a prediction error between the initial prediction and the first noise data to generate a trained prediction model (step S**340**). Specifically, one of multiple targets of the training phase is to minimize a loss function (related to an error/a loss between the output of the prediction model (that is, the initial prediction **422** of FIG. **4**) and the ground truth (for example, the first noise data **412**) in the training sample). In an embodiment, in the training phase of the prediction model, the parameters of the prediction model are recursively updated through minimizing the loss function, for example, by back-propagation. The parameters of the model are, for example, weights, number of layers, positions or number of neurons, activation functions, or offsets, but not limited thereto. The method of

updating the parameters is, for example, through gradient descent, an adaptive moment estimation (Adam) optimizer, momentum, adaptive gradient (Adagrad), or a conjugate gradient method, but not limited thereto. In other words, one of the targets of the training phase is to make the initial prediction output by the prediction model close to or the same as the corresponding ground truth. In an embodiment, the trained prediction model means that the loss function has converged, the prediction accuracy has reached a corresponding threshold, or the training has reached a standard for stopping training early. However, the standard for training completion may still be adjusted according to other tasks or requirements.

[0052] Please refer to FIG. **4**, the loss function is based on a prediction error **440** between the initial prediction **422** and the first noise data **412**. In an embodiment, the prediction error **440** is related to a first error **441** and a second error **442**, such as the sum of the first error **441** and the second error **442**. The processor **120** may calculate a difference between the predicted mean **423** and the mean of the first noise data **412** to determine the first error **441** (for example, an L1 loss). The mean of the first noise data **412** is, for example, the mean of the Gaussian distribution **411**. The first error **441** is, for example, a difference value between the predicted mean **423** and the mean of the Gaussian distribution **411**. The processor **120** may calculate a difference between the predicted variation **424** and the variation of the first noise data **412** to determine the second error **442** (for example, Kullback-Leibler (KL) divergence). The variation of the first noise data **412** is, for example, the variation of the Gaussian distribution **411**. The second error **442** is, for example, a difference value between the predicted variation **424** and the variation of the Gaussian distribution **411**.

[0053] FIG. **5** is a flowchart of spatial noise removal related to a data generation framework according to an embodiment of the disclosure. Please refer to FIG. **5**. The processor **120** inputs second noise data to the trained prediction model, and outputs predicted noise corresponding to the second noise data by referring to second semantic mask data by the trained prediction model (step S**510**). Specifically, the second noise data may be values based on a statistical distribution such as a Gaussian distribution, a uniform distribution, or a Poisson distribution or data generated based on noise in a real environment. It should be noted that to maximize the diversity of generated data, most techniques directly employ the Gaussian distribution, which has the maximum entropy. Other candidate distributions are not as effective in generating diversity, but they may be selected based on specific circumstances. FIG. **6** is a schematic diagram of spatial noise removal related to a data generation framework according to an embodiment of the disclosure. Taking FIG. **6** as an example, the processor **120** samples a Gaussian distribution **611**, and generates second noise data **601** according to sampled values. Next, the processor **120** uses the second noise data **601** as the input of the trained prediction model **430**. That is, the second noise data **601** is input to the trained prediction model **430**.

[0054] The second semantic mask data **603** defines one or more second semantic categories for

[0055] first generated data **625**. The second semantic mask data **603** is marked data with the same size or resolution as the first generated data **625**. Second semantic mask data **603** and **604** are composed of multiple blocks, elements, or pixels. Taking the image form as an example, each pixel in the second semantic mask data **603** and **604** is assigned a predefined (second) semantic category (also referred to as label or semantic information). Taking the scene application as an example, the semantic category may be a lane, a car, a pedestrian, a building, or the sky. Taking a face application as an example, the semantic category may be black skin, a nose, double eyelids, or curly hair. The first generated data **625** is data expected to be generated. The first generated data **625** may be in the image form. In other embodiments, the first generated data **625** may be sound, text, sensing intensity, angle, amplitude, position, or other forms of data.

[0056] In an embodiment, the processor **120** may generate the second semantic mask data **603** through a direct corresponding generator. The generator may generate the second semantic mask data **603** randomly or based on rules. In another embodiment, the processor **120** may receive a user operation, and define one or more second semantic categories in the second semantic mask data **603** and the second semantic category corresponding to the blocks, the elements, or the pixels in the second semantic mask data **603** according to parameters corresponding to the user operation.

[0057] In an embodiment, the processor **120** may generate the second semantic mask data **604** with multiple sizes (or multiple resolutions) (that is, change the size **453** or the resolution of the second semantic mask data **603**). The size or the resolution of the second semantic mask data **604** is smaller or lower than the size or the resolution of one or more second semantic mask data **603**. The second semantic mask data **604** with a certain size is aligned to the size of the feature data (for example, the up-sampled feature map of the output of the encoder **431** or the output of another decoder block **436**) input to the corresponding decoder block **436**, and the second semantic mask data **604** and the feature data with the same size are input to the decoder block **436**. In an embodiment, in response to the decoder **435** including multiple decoder blocks **436**, the second semantic mask data **604** with multiple sizes or resolutions are respectively input to the same or corresponding decoder block **436** to increase the sensitivity of the prediction model **430** to the semantic information with multiple resolutions or sizes.

[0058] The decoder **435** or the trained prediction model **430** outputs predicted noise **621** corresponding to the second noise data **601**. The predicted noise **621** includes a mean and a variation corresponding to the mean. The processor **120** may generate predicted noise **622** based on the predicted noise **621**. For example, the predicted noise **622** is $n+(e^{0.50v}*0.5)$, where n is the mean of the predicted noise **621** and v is the variation of the predicted noise **621**.

[0059] Please refer to FIG. **5**. The processor **120** generates noise removed data according to a difference between the predicted noise and the second noise data (step S**520**). Assuming that data is in the vector form, and the noise removed data is a vector of subtracting the predicted noise from the second noise data. Taking FIG. **6** as an example, the processor **120** subtracts the predicted noise **622** from the second noise data **601**, and generates noise removed data **623**.

[0060] In an embodiment, the processor **120** may input the noise removed data **623** to the trained prediction model **430**, and output the predicted noise corresponding to the noise removed data by referring to the second semantic mask data **604** by the trained prediction model **430**. That is, the processor **120** uses the noise removed data **623** as the input of the trained prediction model **430** or replaces the second

noise data of step S**510** with the noise removed data **623**. Next, the processor **120** performs step S**520**. The processor **120** may repeat the above steps (that is, input the noise removed data **623** to the trained prediction model **430**, output the predicted noise corresponding to the noise removed data by referring to the second semantic mask data **604** by the trained prediction model **430**, and generate another noise removed data **623** according to a difference between the predicted noise **622** and the second noise data **601**) until a stop condition is met. For example, the stop condition is to repeat the above steps 1000 times, but not limited thereto. Noise removed data **624** is noise removed data generated when the stop condition is met.

[0061] Please refer to FIG. **5**. The processor **120** converts the noise removed data into first generated data by a decoder corresponding to the initial encoder of step S**310** (S**530**). In an embodiment, the decoder corresponding to the initial encoder is a vector quantized (VQ) decoder. Taking FIG. **6** as an example, a decoder **651** may convert a discrete latent performance/representation (for example, a (compressed) vector, a code vector, or a codeword form) generated by a vector quantized encoder back into an original data space to implement reconstruction or generation of data. During a decoding process of the decoder **651**, the processor **120** takes the retrieved (compressed) vector, code vector, or codeword as input to be converted back into the original data space through a series of operations (for example, deconvolution, convolution, etc.), to reconstruct the data. For example, the noise removed data **624** is converted into the first generated data **625**. The decoder **651** may map low-dimensional data to a high-dimensional space, and perform feature conversion.

[0062] In another embodiment, the decoder **651** is another decoder for dimensionality enhancement or decompression.

[0063] FIG. **7** is a schematic diagram illustrating noise removal according to an embodiment of the disclosure. Please refer to FIG. **7**. First generated data **701**, **702**, **703**, and **704** are respectively generated data converted through the decoder **651** from the noise removed data **624** generated by repeating the steps (for example, input the noise removed data **623** to the trained prediction model **430**, output the predicted noise **622** corresponding to the noise removed data **623** by referring to the second semantic mask data **604** by the trained prediction model **430**, and generate another noise removed data **623** according to the difference between the predicted noise **622** and the second noise data **601** of FIG. **6**). As the number of repetitions increases (for example, the number of repetitions corresponding to the first generated data **704** is greater than that of the first generated data **703**, the number of repetitions corresponding to the first generated data **703** is greater than that of the first generated data **702**, and the number of repetitions corresponding to the first generated data **702** is greater than that of the first generated data **701**), the image quality gradually improves. In some application scenarios, through repeated noise removal, the quality of data may be effectively improved to be closer to the real situation. For example, the first generated data **704** is a realistic street picture.

[0064] FIG. **8** is a flowchart of multi-domain conditioned learning related to a data generation framework according to an embodiment of the disclosure. Please refer to FIG. **8**. The processor **120** obtains a first code from reference data by a style encoder (step S**810**). Specifically, FIG. **9** is a schematic diagram of multi-domain conditioned learning related to a data generation framework according to an embodiment of the disclosure. Please refer to FIG. **9**. The processor **120** may obtain a training pair (including first source data **901** and reference data **902**) from the storage **110**. The first source data **901** and the reference data **902** may come from different data sets. For example, taking scene generation as an example, the first source data **901** is a street picture of a sunny day, and the reference data **902** is a street picture of a rainy day. However, contents of the first source data **901** and the reference data **902** may still be changed according to actual requirements. The first source data **901** and the reference data **902** may be in the image form. In other embodiments, the first source data **901** and the reference data **902** may be sounds, texts, sensing intensities, angles, amplitudes, positions, or data in other forms.

[0065] A style encoder E includes an encoder E**01**. The encoder E**01** may be a shared encoder for multi-task learning. Multiple tasks share the same encoder E**01**, and representations/performances learned by the encoder E**01** may capture common features between the tasks. The tasks correspond to, for example, different style options E**11**, E**12**, and E**13**. For example, the style option E**11** numbered 0 is a sunny day. The style option E**12** numbered y (y is a positive integer greater than 0) is a rainy day. The style option E**13** numbered $\hat{y}$ is one of the style option E**11** numbered 0 to the style option E**12** numbered y. For example, a rainy day is selected as the style option E**13** numbered $\hat{y}$. The first code is a style code **912** corresponding to the style option E**13**. That is, the encoder E**01** retrieves a feature performance/representation (for example, a vector or a matrix form) generated for the style option E**13** from the reference data **902**.

[0066] In an embodiment, the style encoder E also corresponds to a classifier or a decoder of a task, and the classifier or the decoder is used to convert the output of the encoder E**01** into a prediction result of the corresponding task. For example, a classifier numbered 0 evaluates a generation effect of the sunny day corresponding to the style option E**11** converted from the output of the encoder E**01**. That is, a generation result corresponding to the style option is scored.

[0067] Please refer to FIG. **8**. The processor **120** obtains a second code from latent encoding by a mapping network (step S**820**). Specifically, please refer to FIG. **9**. A mapping network M includes an encoder M**01**. The encoder M**01** may be a shared encoder for multi-task learning. Multiple tasks share the same encoder M**01**, and representations/performances learned by the encoder M**01** may capture common features between the tasks. The tasks correspond to, for example, different style options M**11** and M**13**. For example, the style option M**11** numbered 0 is daytime. The style option M**13** numbered $\hat{y}$ is one of the style option M**11** numbered 0 to style options with other numbers. For example, the daytime is selected as the style option M**13** numbered $\hat{y}$. The second code is a style code **912** corresponding to the style option M**13**. That is, the encoder M**01** retrieves a feature performance/representation (for example, a vector or a matrix form) generated for the style option M**13** from latent encoding **903**.

[0068] In an embodiment, the mapping network M also corresponds to a classifier or a decoder of a task, and the classifier or the decoder is used to convert the output of the encoder M**01** into a prediction result of the corresponding task. For example, the classifier numbered 0 converts the output of the encoder M**01** into an evaluation of a generation

effect of the daytime corresponding to the style option M**11**. That is, a generation result corresponding to the style option is scored.

[0069] In an implementation, the latent encoding **903** is noise data and may be a value based on a statistical distribution such as a Gaussian distribution, a uniform distribution, or a Poisson distribution or data generated based on noise in a real environment. For example, the processor **120** samples the Gaussian distribution, and generates the latent encoding **903** according to sampled values.

[0070] The first code and the second code are both the style codes **912**. The second code is output from the same branch as the style encoder E, that is, respectively fed/input to subsequent modules. In additional, the style code **912** corresponds to one or more style options, such as the style options E**13** and M**13**.

[0071] It should be noted that the contents and the types of the style options may still be changed according to actual requirements and are not limited by the embodiments of the disclosure.

[0072] Please refer to FIG. **8**. The processor **120** inputs first source data to a generator, and outputs a first output corresponding to the first source data by referring to a style code by the generator (step S**830**). Specifically, the first source data is as described above for the first source data of FIG. **9** and will not be repeated here. The generator is a key element in a generative adversarial network (GAN). The generator may generate new data samples (for example, images, text, music, etc.) from a latent space or random noise. In some application scenarios, the generator is used to learn the distribution of real data, and generate the new samples similar to the real data.

[0073] Please refer to FIG. **9**. A generator G includes an encoder G**01** and a decoder G**02**. The encoder G**01** is connected to the decoder G**02**. For example, the output (for example, a content feature **905**/feature map) of the encoder G**01** is used as the input of the decoder G**02**. The encoder G**01** may encode input data (for example, the first source data **901**) into a latent vector (for example, the content feature **905** or a latent feature), and retrieve important information and variation factors in the input data accordingly. The decoder G**02** converts the latent vector into a high-dimensional data representation, and converts the data representation into an output (for example, a first output **921**) in the same format as the real data.

[0074] In an embodiment, the generator G further includes a fusion layer G**03**. The fusion layer G**03** is connected between the encoder G**01** and the decoder G**02**. FIG. **10** is a schematic diagram of the fusion layer G**03** according to an embodiment of the disclosure. Please refer to FIG. **10**. The fusion layer G**03** includes a fusion block **1001** and a predictor **1002**. The fusion block **1001** fuses an output from a certain encoder block (to be introduced in subsequent embodiments) of the encoder G**01** (for example, feature data **111** or other feature representations/expressions, and the fusion block **1001** of the encoder block is connected through a residual connection) and the style code **912** from the style encoder E and/or the mapping network M to generate fusion data, and enhance style information in the output of the certain decoder block of the encoder G**01** accordingly. The predictor **1002** is connected to the fusion block **1001**. The predictor **1002** may convert the fusion data output by the fusion block **1001** into the style option corresponding to the style code **912**. The output of the fusion layer G**03** may be used as an input **112** of the certain decoder block (to be introduced in subsequent embodiments) of the decoder G**02**. That is, the processing of the reference style code **912** is implemented through the fusion processing of the fusion layer G**03**.

[0075] FIG. **11** is a schematic diagram of the generator G according to an embodiment of the disclosure. Please refer to FIG. **11**. The encoder G**01** includes one or more encoder blocks G**11** (for example, a block numbered 1, a block numbered 2 to a block numbered N, where N is a positive integer). The encoder block G**11** includes, for example, a convolutional layer, down-sampling processing, and residual connection processing, but not limited thereto. The decoder G**02** includes one or more decoder blocks G**12** (for example, a block numbered 1, a block numbered 2 to a block numbered N, where N is a positive integer). The decoder block G**12** includes, for example, a deconvolutional layer, a fully connected layer, up-sampling processing, and excitation function processing, but not limited thereto. The fusion layer G**03** may be inserted into a U-net similar network. For example, the encoder G**01** shown in the drawing is composed of the encoder G**01** and the decoder G**02**, the encoder G**01** may down-sample the dimension of the latent space, and the decoder G**02** may up-sample the dimension of the latent space. The generator G may include one or more fusion layers G**03**. The input of each fusion layer G**03** is the same as or corresponds to the size or the resolution of the output. The processor **120** may input the fusion data and another feature data to the decoder block G**12**, and the another feature data is the output (for example, the content feature **905**) of the encoder block G**11** located adjacent to the front (for example, adjacent to the left in the drawing) or the output of another decoder block G**12** located adjacent to the front (for example, adjacent to the left in the drawing).

[0076] In addition, the user may insert one or more fusion layers G**03** between the encoder block G**11** and the decoder block G**12** corresponding to a specific size or dimension according to actual requirements, which are not limited by the embodiments of the disclosure.

[0077] Since the decoder G**02** also inputs the output of the fusion layer G**03** (the style information entrained with the style code **912**), the output (for example, the first output **921** of FIG. **9**) of the decoder G**02** may be regarded as style translation data. Taking FIG. **9** as an example, the first source data **901** is in an image format, and the first output **921** is a style translation image. In addition, the first output **921** has the style option (taking the rainy day as an example) corresponding to the style code **912**, but the same lane, vehicle, and scenery in the first source data **901** are still retained.

[0078] Please refer to FIG. **8**. The processor **120** inputs the first output to a discriminator, and outputs a second output corresponding to the first output by the discriminator (step D**01**). Specifically, the discriminator is another key element in the generative adversarial network (GAN). The generator may distinguish between real data (for example, images, text, music, etc.) and the first output that is output by the generator G, that is, judge whether the input data (for example, the first output of the generator) is true (real) or false (fake). In an embodiment of the disclosure, the second output that is output by the discriminator includes real or fake corresponding to the style option, that is, the second output represents whether the first output corresponds to the style option. "Real" means corresponding to or the same as

the style option, and "fake" means not corresponding to or different from the style option. The discriminator includes, for example, a convolutional layer, a fully connected layer, an activation function, and a normalization layer, but not limited thereto.

[0079] Taking FIG. **9** as an example, a discriminator D includes an encoder D**01**. The encoder D**01** may be a shared encoder for multi-task learning. Multiple tasks share the same encoder D**01**, and representations/performances learned by the encoder D**01** may capture common features between the tasks. The tasks correspond to, for example, different style options. For example, the style option numbered 0 is the sunny day. The style option numbered y (y is a positive integer greater than 0) is the rainy day. The style option numbered y is one of the style option numbered 0 to the style option numbered y. For example, the rainy day is selected as the style option numbered ŷ. The encoder D**01** may retrieve a feature performance/representation (for example, a vector or a matrix form) generated for a specific style option from the first output **921**.

[0080] The discriminator D includes a predictor D**11** numbered 0 to a predictor D**12** numbered y (y is a positive integer). A predictor D**13** numbered ŷ is one of the predictor D**11** numbered 0 to the predictor D**12** numbered y. Each numbered predictor is used to judge whether a specific style option is corresponded to. For example, the predictor D**11** numbered 0 judges whether there is the style option of the sunny day. "Real" means corresponding to or the same as the style of the sunny day, and "fake" means not corresponding to or different from the style of the sunny day.

[0081] In an embodiment, the style options corresponding to the predictors D**11** to D**13** of the discriminator D may correspond to the style options of the style encoder E and the mapping network M. In other embodiments, contents of the style options may still be changed according to actual requirements.

[0082] In an embodiment, please refer to FIG. **9**. The processor **120** may update at least one of the style encoder E, the mapping network M, the generator G, and the discriminator D according to a style error **941**. One of the targets of the training phase is to minimize the loss function (related to the error/loss between the output of the style encoder E, the mapping network M, the generator G, and/or the discriminator D and the corresponding data). In an embodiment, in the training phase of the prediction model, the parameters of the prediction model are recursively updated through minimizing the loss function, for example, by back-propagation. The parameters of the model are, for example, weights, number of layers, positions or number of neurons, activation functions, or offsets, but not limited thereto. The method of updating the parameters is, for example, through gradient descent, an adaptive moment estimation optimizer, momentum, adaptive gradient, or a conjugate gradient method, but not limited thereto. In other words, one of the targets of the training phase is to make it difficult for the discriminator D to identify whether the output of the generator G is "real" or "fake". In an embodiment, the trained prediction model means that the loss function has converged, the prediction accuracy has reached the corresponding threshold, or the training has reached the standard for stopping training early. However, the standard for training completion may still be adjusted according to other tasks or requirements.

[0083] FIG. **12**A is a schematic diagram of a style similarity error L_{SS} according to an embodiment of the disclosure. Please refer to FIG. **12**A. The style error **941** of FIG. **9** includes the style similarity error L_{SS}. The processor **120** may calculate a difference between first codes $\tilde{s}_1$ and $\tilde{s}_2$ (forming a positive pair PP**1**) respectively obtained from first source data **121** (for example, the first source data **901** of FIG. **9**) and second source data **122** by the style encoder E to determine the style similarity error L_{SS}. The second source data **122** and the first source data **121** correspond to the same style option, such as the sunny day. In addition, the style similarity error L_{SS} is, for example, a difference value obtained by subtracting the first code $\tilde{s}_2$ from the first code $\tilde{s}_1$ in the vector form. In other words, one of the targets of the loss function is that the style codes generated by the style encoder E for multiple data of the same style option are the same or similar.

[0084] FIG. **12**B is a schematic diagram of a source preservation error L_{SP} according to an embodiment of the disclosure. Please refer to FIG. **12**B. The style error **941** of FIG. **9** includes the source preservation error L_{SP}. The processor **120** may calculate a difference between first source data **123** (for example, the first source data **901** of FIG. **9**) and second generated data **127** to determine the source preservation error L_{SP}. The generator G refers to a style code **9121** and outputs third generated data **124** corresponding to the first source data **123**. The style code **9121** is a first code s̃ obtained from reference data **125** (for example, the reference data **902** of FIG. **9**) by the style encoder E. The generator G refers to a second style code **9122** and outputs the second generated data **127** corresponding to the third generated data **124**. The second style code **9122** is obtained from second reference data **126** by the style encoder E, and the second reference data **126** and the first source data **123** correspond to the same style option (for example, the sunny day). The reference data **125** and the first source data **123** may correspond to the same or different style options. In addition, the source preservation error L_{SP} is, for example, a difference value obtained by subtracting the second generated data **127** from the first source data **123** in the matrix form. In other words, one of the targets of the loss function is that the second generated data **124** generated by the generator G may be restored to the first source data **123** or similar to the first source data **123** by referring to the style code of the same style option as the first source data **123** through the generator G.

[0085] FIG. **12**C is a schematic diagram of a content comparison error L_{CC} according to an embodiment of the disclosure. Please refer to FIG. **12**C. The style error **941** of FIG. **9** includes the content comparison error L_{CC}. The processor **120** may calculate a difference between a first feature representation **9051** and a second feature representation **9052** (forming a positive pair PP**2**), and calculate a difference between the second feature representation **9052** and a third feature representation **9053** (forming a negative pair NP**2**)) to determine the content comparison error L_{cc}. The first feature representation **9051** is obtained from first source data **128** (for example, the first source data **901** of FIG. **9**) by the encoder G**01** of the generator G. The generator G refers to a style code **9123** and outputs third generated data **130** corresponding to the first source data **128**. The style code **9123** is the first code s obtained from reference data **129** (for example, the reference data **902** of FIG. **9**) by the style encoder E. The second feature repre-

sentation **9052** is obtained from the third generated data **130** by the encoder G**01**. In addition, the third feature representation **9053** is obtained from reference data **129** by encoder G**01**. The content comparison error L_{CC} uses, for example, the difference (for example, a difference value) between the second feature representation **9052** and the third feature representation **9053** as the denominator and the difference (for example, a difference value) between the first feature representation **9051** and the second feature representation **9052** as the numerator. Since one of the targets of the loss function is to minimize the output value, the larger the difference (considered as content similarity) between the second feature representation **9052** and the third feature representation **9053** the better and/or the smaller the difference (considered as content similarity) between the first feature representation **9051** and the second feature representation **9052** the better. Since the encoder GO**1** is used to retrieve the content feature, the more similar or the closer the first source data **128** and the content feature of the third generated data **130** generated by the generator G the better, but the less similar or the less close the reference data **129** and the content feature of the third generated data **130** the better.

[0086] In an embodiment, the style error **941** of FIG. **9** may be a sum of the style similarity error L_{SS} of FIG. **12**A, the source preservation error L_{SP} of FIG. **12**B, and the content comparison error L_{CC} of FIG. **12**C. In another embodiment, depending on different design requirements, the style error **941** may also be a result of weighted operations of the style similarity error L_{SS}, the source preservation error L_{SP}, and the content comparison error L_{CC}. That is, corresponding weights or priorities are respectively given to the style similarity error L_{SS}, the source preservation error L_{SP}, and the content comparison error L_{CC}.

[0087] FIG. **13** is a schematic diagram of a style option according to an embodiment of the disclosure. Please refer to FIG. **13**. Different style options may be customized according to different design requirements. Taking scenario simulation as an example, the style options of a weather condition are sunny day, rainy day, cloudy day, etc.; the style options of a day/night condition are day and night; and the style options of a scene condition are city, street, expressway, etc. The processor **120** may respectively train module pairs (that is, the style encoder E and the mapping network M corresponding to the same condition/type). The user may select the required condition/type according to requirements.

[0088] FIG. **14** is a flowchart of inference of multi-domain conditions related to a data generation framework according to an embodiment of the disclosure. Please refer to FIG. **14**. The processor **120** obtains a third code from second latent encoding by the (trained) mapping network (step S**1401**). Specifically, the second latent encoding may be a value based on a statistical distribution such as a Gaussian distribution, a uniform distribution, or a Poisson distribution or data generated based on noise in a real environment. FIG. **15** is a schematic diagram of inference of multi-domain conditions related to a data generation framework according to an embodiment of the disclosure. Taking FIG. **15** as an example, the processor **120** obtains the first generated data or the third source data generated in step S**530** of FIG. **5** from the storage **110**. Reference may be made to the first source data **901** of FIG. **9** for the introduction of third source data **1501**, which will not be repeated here.

[0089] In addition, the processor **120** samples a Gaussian distribution **1511**, and generates second latent encoding **1502** according to sampled values. Next, the processor **120** takes the second latent encoding **1502** as the input of the trained mapping network M. That is, the second latent encoding **1502** is input to the trained mapping network M. Representations/performances learned by the encoder M**01** of the mapping network M may capture common features between the tasks. The tasks correspond to, for example, different style options M**11**, M**12**, and M**13**. For example, the style option M**11** numbered 0 is the cloudy day, the style option M**12** numbered y is the sunny day, and the style options M**13** numbered $\hat{y}$ is the rainy day. The style option M**13** numbered $\hat{y}$ is one of the style option M**11** numbered 0 to style options with other numbers. As mentioned above for the introduction of the second code, the second code generated by the mapping network M is a style code corresponding to a certain style option. Similarly, the third code generated by the mapping network M is a style code **1512** corresponding to a certain style option, such as corresponding to the style option M**13** numbered $\hat{y}$.

[0090] Please refer to FIG. **14**. The processor **120** inputs the first generated data or the third source data to the trained generator, and outputs fourth generated data corresponding to the first generated data or the third source data by referring to the third code by the trained generator (step S**1402**). Please refer to FIG. **15**. The encoder G**01** of the generator G retrieves a content feature **1505** from the first generated data or the third source data **1501**. The fusion layer G**03** fuses the feature data and the style code **1512** output by a certain encoder block (for example, the encoder block G**11** of FIG. **11**) of the encoder G**01**, and generates the fusion data. The processor **120** inputs the fusion data and the feature data (for example, the content feature **1505** or the output of other decoder blocks) to the corresponding decoder block of the decoder G**02**. Finally, the decoder G**02** outputs fourth generated data **1503**. The fourth generated data **1503** is data with the same content as the third source data **1501** or the first generated data and the style option corresponding to the style code **1502**.

[0091] FIG. **16**A is a schematic diagram of scene image generation according to an embodiment of the disclosure. Please refer to FIG. **16**A. The embodiment of the disclosure provides a spatial conditioned pipeline and a multi-domain conditioned pipeline. For the application scenario of scene image generation (for example, a scene for self-driving car training), a simulator (for example, a car learning to act (CARLA) simulator or an annotated view sequence) may generate semantic mask data **1601**. The semantic mask data **1601** defines semantic categories corresponding to multiple pixels of generated data **1602**. For example, the pixels in a certain image region correspond to a lane, the pixels in another image region correspond to a vehicle, and the pixels in another image region correspond to the sky. It should be noted that the semantic mask data **1601** is not limited to defining two-dimensional data, but may also be used to define three-dimensional, four-dimensional, or more dimensional data. For example, a three-dimensional semantic mask is composed of three channels (that is, height, width, and number of categories), and a four-dimensional semantic mask further includes a time dimension compared to the three-dimensional semantic mask. A data generator **1611** (used to execute step S**510** to step S**530** of FIG. **5**) may refer to the semantic mask data **1601** and output the predicted

noise, generate the noise removed data according to the difference between the predicted noise and the noise data, and convert the noise removed data into the generated data **1602** by the decoder corresponding to the initial encoder. Multiple image regions in the generated data **1602** may correspond to the corresponding semantic categories defined by the semantic mask data **1601**.

[0092] Next, a condition fuser **1612** (used to execute step S**1401** and step S**1402** of FIG. **14**) obtains a style code corresponding to a style option **1631** (for example, a snowy day or a rainy day), and converts the input generated data **1602** into generated data **1603**, such as a street picture of the rainy day, by the trained generator. The processor **120** may store the generated data **1603** in a database corresponding to the style option **1631**. Multiple generated data **1603** may form a data set. The style option **1631** may be an input content corresponding to a user operation or may be generated based on specific conditions or random numbers and may be changed according to requirements of the user.

[0093] FIG. **16**B is a schematic diagram of facial image generation according to an embodiment of the disclosure. Please refer to FIG. **16**B. For the application scenario of facial image generation, semantic mask data **1604** may be obtained from the data set. The semantic mask data **1604** defines semantic categories corresponding to multiple pixels of the generated data **1605**. For example, the pixels in a certain image region correspond to hair, the pixels in another image region correspond to a nose, and the pixels in another image region correspond to a mouth. Similarly, the data generator **1611** (used to execute step S**510** to step S**530** of FIG. **5**) may refer to the semantic mask data **1604** and output the predicted noise, generate the noise removed data according to the difference between the predicted noise and the noise data, and convert the noise removed data into the generated data **1605** by the decoder corresponding to the initial encoder. Multiple image regions in the generated data **1605** may correspond to the corresponding semantic categories defined by the semantic mask data **1604**.

[0094] Next, the condition fuser **1612** (used to execute step S**1401** and step S**1402** of FIG. **14**) obtains a style code corresponding to a style option **1632** (for example, a male or a female), and converts the input generated data **1605** into generated data **1606**, such as a facial picture of the female, by the trained generator.

[0095] It should be noted that the training data sets for the spatial conditioned pipeline and the multi-domain conditioned pipeline may be different, and the training stages of the two pipelines may be performed respectively.

[0096] FIG. **17**A and FIG. **17**B are experimental results of a spatial conditioned pipeline according to an embodiment of the disclosure. Please refer to FIG. **17**A. Generated data **1703** corresponding to a ground truth **1702** may be generated based on semantic mask data **1701**. Multiple image regions in the generated data **1703** may correspond to corresponding semantic categories defined by the semantic mask data **1701**. Please refer to FIG. **17**B. Generated data **1706** corresponding to a ground truth **1705** may be generated based on semantic mask data **1704**. Multiple image regions in the generated data **1706** may correspond to corresponding semantic categories defined by the semantic mask data **1704**.

[0097] FIG. **18** is an experimental result of a multi-domain conditioned pipeline according to an embodiment of the disclosure. Please refer to FIG. **18**. Assuming that an input image is a street picture of a sunny day, street pictures with style options of a cloudy day, a rainy day, and a snowy day may be generated respectively. Except for weather, image contents of the street pictures all correspond to the street picture of the input image. Alternatively, assuming that input images are respectively street pictures of the snowy day, the rainy day, and the cloudy day, the street pictures may be respectively transformed into the street picture of the sunny day, but the original street content is still retained.

[0098] In summary, in the method and the apparatus related to the data generation framework according to the embodiments of the disclosure, the prediction model for identifying/predicting the noise data is trained, and the corresponding noise removed data is converted into the generated data. In this way, the high-resolution data that is close to the real world may be generated. In addition, the style code corresponding to the style option is retrieved, the generator refers to the style code, and judges whether the output of the generator is real or fake through the discriminator. In this way, the trained generator may be used to generate the generated data that conforms to the specific style option, and the generated data may still retain the source content. The generated data may be evaluated on various data sets to verify the quality and the robustness of the data generation.

[0099] Although the disclosure has been disclosed in the above embodiments, the embodiments are not intended to limit the disclosure. Persons skilled in the art may make some changes and modifications without departing from the spirit and scope of the disclosure. Therefore, the protection scope of the disclosure shall be defined by the appended claims.

What is claimed is:

1. A method related to a data generation framework, comprising:

obtaining a first code from reference data by a style encoder;

obtaining a second code from latent encoding by a mapping network, wherein the first code and the second code are both a style code, and the style code corresponds to at least one style option;

inputting first source data to a generator, and outputting a first output corresponding to the first source data by referring to the style code by the generator; and

inputting the first output to a discriminator, and outputting a second output corresponding to the first output by the discriminator, wherein the second output represents whether the first output corresponds to the at least one style option.

2. The method related to the data generation framework according to claim **1**, wherein the generator comprises a first encoder and a first decoder, the first encoder is connected to the first decoder, the first encoder comprises at least one first encoder block, the first decoder comprises at least one first decoder block, and the step of outputting the first output corresponding to the first source data by referring to the style code by the generator comprises:

fusing feature data output by one of the at least one first encoder block and the style code to generate fusion data; and

inputting the fusion data and another feature data to one of the at least one first decoder block, wherein the another feature data is an output of one of the at least

one first encoder block or an output of another one of the at least one first decoder block.

3. The method related to the data generation framework according to claim **1**, further comprising:

updating at least one of the style encoder, the mapping network, the generator, and the discriminator according to a style error, wherein the style error comprises a style similarity error, and the step of updating the at least one of the style encoder, the mapping network, the generator, and the discriminator according to the style error comprises:

calculating a difference between the two first codes respectively obtained from the first source data and second source data by the style encoder to determine the style similarity error.

4. The method related to the data generation framework according to claim **1**, further comprising:

updating at least one of the style encoder, the mapping network, the generator, and the discriminator according to a style error, wherein the style error comprises a source preservation error, and the step of updating the at least one of the style encoder, the mapping network, the generator, and the discriminator according to the style error comprises:

calculating a difference between the first source data and second generated data to determine the source preservation error, wherein the generator outputs third generated data corresponding to the first source data by referring to the style code, the generator outputs the second generated data corresponding to the third generated data by referring to a second style code, the second style code is obtained from second reference data by the style encoder, and the second reference data and the first source data correspond to a same one of the at least one style option.

5. The method related to the data generation framework according to claim **1**, further comprising:

updating at least one of the style encoder, the mapping network, the generator, and the discriminator according to a style error, wherein the style error comprises a content comparison error, and the step of updating the at least one of the style encoder, the mapping network, the generator, and the discriminator according to the style error comprises:

calculating a difference between a first feature representation and a second feature representation, and calculating a difference between the second feature representation and a third feature representation to determine the content comparison error, wherein the first feature representation is obtained from the first source data by a first encoder of the generator, the generator outputs third generated data corresponding to the first source data by referring to the style code, the second feature representation is obtained from the third generated data by the first encoder, and the third feature representation is obtained from the reference data by the first encoder.

6. The method related to the data generation framework according to claim **1**, further comprising:

obtaining a third code from second latent encoding by the mapping network; and

inputting first generated data or third source data to the trained generator, and outputting fourth generated data corresponding to the first generated data or the third source data by referring to the third code by the trained generator.

7. The method related to the data generation framework according to claim **1**, further comprising:

obtaining a latent representation from training data by an initial encoder;

combining the latent representation and first noise data to generate a noisy latent representation;

inputting the noisy latent representation to a prediction model, and outputting an initial prediction corresponding to the noisy latent representation by referring to first semantic mask data by the prediction model, wherein the first semantic mask data defines at least one first semantic category for the training data; and

updating the prediction model according to a prediction error between the initial prediction and the first noise data to generate a trained prediction model.

8. The method related to the data generation framework according to claim **7**, wherein the prediction model comprises a second encoder and a second decoder, the second encoder is connected to the second decoder, the second encoder comprises at least one second encoder block, the second decoder comprises at least one second decoder block, and the step of outputting the initial prediction corresponding to the noisy latent representation by referring to the first semantic mask data by the prediction model comprises:

inputting the noisy latent representation to the second encoder;

inputting the first semantic mask data and feature data to one of the at least one second decoder block, wherein the feature data is an output of one of the at least one second encoder block or an output of another one of the at least one second decoder block; and

outputting the initial prediction by the second decoder.

9. The method related to the data generation framework according to claim **7**, wherein the prediction error is a sum of a first error and a second error, the initial prediction comprises a predicted mean and a predicted variation, and the step of updating the prediction model according to the prediction error between the initial prediction and the first noise data comprises:

calculating a difference between the predicted mean and a mean of the first noise data to determine the first error; and

calculating a difference between the predicted variation and a variation of the first noise data to determine the second error.

10. The method related to the data generation framework according to claim **7**, further comprising:

inputting second noise data to the trained prediction model, and outputting predicted noise corresponding to the second noise data by referring to second semantic mask data by the trained prediction model, wherein the second semantic mask data defines at least one second semantic category for first generated data;

generating noise removed data according to a difference between the predicted noise and the second noise data; and

converting the noise removed data into the first generated data by a decoder corresponding to the initial encoder.

11. An apparatus related to a data generation framework, comprising:

a storage, used to store a program code; and

a processor, coupled to the storage and configured to load the program code to execute:

obtaining a first code from reference data by a style encoder;

obtaining a second code from latent encoding by a mapping network, wherein the first code and the second code are both a style code, and the style code corresponds to at least one style option;

inputting first source data to a generator, and outputting a first output corresponding to the first source data by referring to the style code by the generator; and

inputting the first output to a discriminator, and outputting a second output corresponding to the first output by the discriminator, wherein the second output represents whether the first output corresponds to the at least one style option.

12. The apparatus related to the data generation framework according to claim **11**, wherein the prediction model comprises a first encoder and a first decoder, the first encoder is connected to the first decoder, the first encoder comprises at least one first encoder block, the first decoder comprises at least one first decoder block, and the processor is further configured to:

fuse feature data output by one of the at least one first encoder block and the style code to generate fusion data; and

input the fusion data and another feature data to one of the at least one first decoder block, wherein the another feature data is an output of one of the at least one first encoder block or an output of another one of the at least one first decoder block.

13. The apparatus related to the data generation framework according to claim **11**, wherein the processor is further configured to:

update at least one of the style encoder, the mapping network, the generator, and the discriminator according to a style error, wherein the style error comprises a style similarity error, and the step of updating the at least one of the style encoder, the mapping network, the generator, and the discriminator according to the style error comprises:

calculating a difference between the two first codes respectively obtained from the first source data and second source data by the style encoder to determine the style similarity error.

14. The apparatus related to the data generation framework according to claim **11**, wherein the processor is further configured to:

update at least one of the style encoder, the mapping network, the generator, and the discriminator according to a style error, wherein the style error comprises a source preservation error, and the step of updating the at least one of the style encoder, the mapping network, the generator, and the discriminator according to the style error comprises:

calculating a difference between the first source data and second generated data to determine the source preservation error, wherein the generator outputs third generated data corresponding to the first source data by referring to the style code, the generator outputs the second generated data corresponding to the third generated data by referring to a second style code, the second style code is obtained from second reference data by the style encoder, and the second reference data and the first source data correspond to a same one of the at least one style option.

15. The apparatus related to the data generation framework according to claim **11**, wherein the processor is further configured to:

update at least one of the style encoder, the mapping network, the generator, and the discriminator according to a style error, wherein the style error comprises a content comparison error, and the step of updating the at least one of the style encoder, the mapping network, the generator, and the discriminator according to the style error comprises:

calculating a difference between a first feature representation and a second feature representation, and calculating a difference between the second feature representation and a third feature representation to determine the content comparison error, wherein the first feature representation is obtained from the first source data by a first encoder of the generator, the generator outputs third generated data corresponding to the first source data by referring to the style code, the second feature representation is obtained from the third generated data by the first encoder, and the third feature representation is obtained from the reference data by the first encoder.

16. The apparatus related to the data generation framework according to claim **11**, wherein the processor is further configured to:

obtain a third code from second latent encoding by the mapping network; and

input first generated data or third source data to the trained generator, and outputting fourth generated data corresponding to the first generated data or the third source data by referring to the third code by the trained generator.

17. The apparatus related to the data generation framework according to claim **11**, wherein the processor is further configured to:

obtain a latent representation from training data by an initial encoder;

combine the latent representation and first noise data to generate a noisy latent representation;

input the noisy latent representation to a prediction model, and output an initial prediction corresponding to the noisy latent representation by referring to first semantic mask data by the prediction model, wherein the first semantic mask data defines at least one first semantic category for the training data; and

update the prediction model according to a prediction error between the initial prediction and the first noise data to generate a trained prediction model.

18. The apparatus related to the data generation framework according to claim **17**, wherein the prediction model comprises a second encoder and a second decoder, the second encoder is connected to the second decoder, the second encoder comprises at least one second encoder block, the second decoder comprises at least one second decoder block, and the processor is further configured to:

input the noisy latent representation to the second encoder;

input the first semantic mask data and feature data to one of the at least one second decoder block, wherein the feature data is an output of one of the at least one

second encoder block or an output of another one of the at least one second decoder block; and

outputting the initial prediction by the second decoder.

19. The apparatus related to the data generation framework according to claim **17**, wherein the prediction error is a sum of a first error and a second error, the initial prediction comprises a predicted mean and a predicted variation, and the processor is further configured to:

calculate a difference between the predicted mean and a mean of the first noise data to determine the first error; and

calculate a difference between the predicted variation and a variation of the first noise data to determine the second error.

20. The apparatus related to the data generation framework according to claim **17**, wherein the processor is further configured to:

input second noise data to the trained prediction model, and output predicted noise corresponding to the second noise data by referring to second semantic mask data by the trained prediction model, wherein the second semantic mask data defines at least one second semantic category for first generated data;

generate noise removed data according to a difference between the predicted noise and the second noise data; and

convert the noise removed data into the first generated data by a decoder corresponding to the initial encoder.

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